

Version 1.2.5 of API contract is [here](#)

Version History

Version	Date	Changes
1.2	15th May 2024	a. Added 2 new fulfillment states "At-pickup", "At-delivery", details here ;
	7th Apr 2024	a. Removed "Backward Compatibility" section;
	23rd Dec 2023	a. Linking cancellation with IGM, pre-cancel state;
	18th Nov 2023	a. Added support for reverse qc codification; b. Added support for communication of fulfillment delay (pickup / delivery);
	4th Nov 2023	a. Added tracking attributes in /on_status to optimize track calls for hyperlocal / hub-based tracking;
	2nd Nov 2023	a. Updated backward compatibility requirements , for pre-order & post-order flows, so that correct version is used;
	7th Oct 2023	a. Updated flow for /confirm & /on_confirm to address dangling orders; b. Additions to /on_track to support non-hyperlocal orders (in addition to tracking of hyperlocal orders already enabled);
	21st Sep 2023	a. Added publishing static terms for transaction level contract in /search; b. Added backward compatibility requirements ;
	19th July 2023	a. Changed /track & /on_track - removed callback_url, updates will be through polling only; b. Updated swagger link;
	20th June 2023	a. Added to /on_update - proof of weight differential, separated updated dimensions & weight (in case of weight differential) from original dimensions & weight;
	8th June 2023	a. Added features as per the feature doc (changes from v1.1 highlighted)

1. Introduction

ONDC API contracts map business requirements for various use case scenarios to a set of attribute keys in different APIs and ensures interoperability, between a buyer NP & seller NP, as follows:

- by defining the set of attribute keys that need to be exchanged to establish a handshake between the buyer NP & seller NP;
- clearly defining the single source of truth for each attribute key and thereby establishing mutability & immutability of these keys on behalf of the participants at each end of the transaction;

This API contract document is structured as follows:

- JSON payload for each API request & response;
- Definition & enumerations of specific attribute keys, as applicable;
- Examples of usage of specific attribute keys for a business requirement, as applicable;
- Expected behaviour of APIs on the network;

Logistics API contract, defines the interaction between the logistics buyer NP (currently, same as retail seller NP which fills the order) and the logistics seller NP (LSP) which fulfills the order, and includes the following APIs:

a. **Pre-order APIs**

- **/search** - buyer NP specifies the search intent;
- **/on_search** - seller NP responds with the catalog based on the search intent;
- **/init & /on_init** - buyer & seller NP specify & agree to the terms & conditions prior to placing the order;
- **/confirm** - buyer NP places the order on behalf of the buyer;
- **/on_confirm** - seller NP responds to the order placed either through auto-acceptance or deferred acceptance or rejection of the order;

b. **Post-order APIs**

- **/status** - buyer NP requests for the current status of the order;
- **/on_status** - seller NP provides the current status of the order;
- **/cancel** - buyer NP places cancellation request for the order;
- **/on_cancel** - seller NP responds to the buyer NP cancellation request or cancels the order directly;
- **/update** - buyer NP updates fulfillment details, such as PCC / DCC / authorization codes / payload details;
- **/on_update** - seller NP responds to the buyer NP request and provides information related to pickup / delivery time slots, shipping label / AWB no / EBN;
- **/track** - buyer NP requests for live tracking of order;
- **/on_track** - seller NP response with URL for live tracking of order;

All attribute keys, defined in this API contract, are mandatory, unless explicitly mentioned otherwise. Participants may validate the request & response payloads, based on mandatory attribute keys, as defined in this contract and respond with "NACK" (along with appropriate error code as defined [here](#)) for invalid payloads. Any payload, without optional attribute keys, cannot be considered as invalid.

All participants are required to ensure full compliance with this API contract on the live ONDC network.

2. Signing & Verification of requests & responses

- a. Key pairs, for signing & encryption, can be generated using standard libraries such as [libsodium](#).
- b. Creating key pairs
 - Create key pairs, for signing (ed25519) & encryption (X25519);
 - Update base64 encoded public keys in registry;
 - Utility to generate signing key pairs and test signing & verification is [here](#);
- c. Auth Header Signing
 - Generate UTF-8 byte array from json payload;
 - Generate Blake2b hash from UTF-8 byte array;
 - Create base64 encoding of Blake2b hash, this becomes the digest for signing;
 - Sign the request, using your private signing key, and add the signature to the request authorization header, following steps documented [here](#);
- d. Auth Header Verification
 - Extract the digest from the encoded signature in the request;
 - Get the signing_public_key from registry using lookup (by using the ukId in the authorization header);
 - Create (UTF-8) byte array from the **raw payload** and generate Blake2b hash;
 - Compare generated Blake2b hash with the decoded digest from the signature in the request;
 - In case of failure to verify, HTTP error 401 should be thrown;

3. Registry lookup for verifying requests

- a. There are 2 possible participant types, for logistics domain, in the ONDC registry:
 - Buyer NP
 - Seller NP
- b. Each participant type, defined above, will have an entry in the registry for their entity. Every participant on the ONDC registry will have 1 or more entries in the registry, with "ukid" for each such entry;
- c. Signing & Verification of requests and responses using auth headers is defined [here](#);
- d. All requests & responses have to be signed by the sender and verified by the receiver. If any request fails verification, it should be rejected (NACK) with http 401 error code;
- e. The identity of the sender of a request is sent, through the auth header, as follows:

```
keyId="{subscriber_id}|{unique_key_id}|{algorithm}"
```

- f. Signing by Buyer App & Seller App will be using their private key, for which the corresponding public key will be in the registry for their "ukid";
- g. Verification of requests using registry /lookup uses [this](#) API spec. **Participants can either use the registry lookup endpoints from [here](#) or cache the registry locally (with refresh at regular interval) for a local lookup.** Following options are available:

- **/lookup**

request:

```
{
  "subscriber_id": "lsp.com",
  "domain": "nic2004:60232",
  "ukid": "UKID1",
  "country1": "IND",
  "city2": "std:080",
  "type3": "BPP"
}
```

response:

(will be an array with 1 or more objects, depending on whether the entity, identified by the subscriber_id, has registered as different participants in a domain, e.g. buyer app, seller app)

```
[
  {
    "subscriber_id": "logistics_buyer.com",
    "status": "SUBSCRIBED",
    "ukid": "UKID1",
    "subscriber_url": "https://logistics_buyer.com/ondc",
    "country": "IND",
    "domain": "nic2004:60232",
    "valid_from": "2023-05-23T00:00:00.000Z",
    "valid_until": "2027-05-23T00:00:00.000Z",
  }
]
```

¹ optional

² optional

³ optional, enum - "BPP", "BAP", "BG"

```

    "type": "BPP",
    "signing_public_key":
    "9V6WbbMwWy953zUIIICOOQq4nk8zHHJRhrZ19juApL4=",
    "encr_public_key":
    "MCowBQYDK2VuAyEATDpic13936lmOrDdzMpQox0KWXMwb9sqsd6fcD1LHM=",
    "created": "2023-05-23T00:00:00.000Z",
    "updated": "2023-05-23T00:00:00.000Z",
    "br_id": "br_id1",
    "city": "std:080"
  }
]

```

○ **/vlookup**

```

{
  "sender_subscriber_id4": "logistics_buyer.com",
  "request_id5": "123456789",
  "timestamp6": "2023-01-26T19:00:00.000Z",
  "search_parameters":
  {
    "domain": "nic2004:60232",
    "subscriber_id7": "lsp.com",
    "country": "IND",
    "type": "sellerApp8",
    "city": "std:080"
  },
  "signature9": ""
}

```

⁴ subscriber id of request initiator

⁵ unique identifier for request

⁶ timestamp in RFC3339 format

⁷ optional

⁸ enums - "buyerApp", "sellerApp", "gateway"

⁹ search_parameters signed using private key of request initiator:

sign(country|domain|type|city|subscriber_id) => sign(IND|nic2004:60232|sellerApp|std:080|lsp.com)

4. Construct of APIs & transaction trail

- a. All APIs follow the same construct and include:
 - **Context** - includes transaction_id, message_id and other attribute keys that identify the source & destination of the message being sent through the API request or response;
 - **Message** - identifies the content of the message e.g. the search intent, catalog details, order details, etc.;
- b. All APIs are asynchronous, i.e.:
 - request is sent from sender to recipient;
 - recipient acknowledges the request by sending an "ACK" or sends "NACK" if the recipient cannot validate the request;
 - recipient sends response through the callback for the corresponding request API e.g. /on_search callback for /search request;
 - recipient of callback (could be the original sender) acknowledges the response by sending an "ACK" or "NACK" if the recipient cannot validate the request;
- c. Every request & response pair is identified by a transaction_id + unique message_id & the sender of a request is able to correlate the response to the request using the above;
- d. If a stale request or response is received, i.e. with timestamp earlier than similar request or response (identified by the transaction_id + message_id) that has already been processed, the recipient can send "NACK" with the error code "65003";
- e. Error block can be sent in sync (in response to request) or async (along with callback) mode and should have the following attribute keys: error.type, error.code;
- f. ONDC standard error codes are defined [here](#);
- g. Transaction trail for a buyer is maintained through a unique transaction_id for the pre-order stage, i.e. /search, /select, /init, /confirm;
- h. Unique id for an order on a network will be identified by the transaction_id & order id;
- i. Swagger link is [here](#);

5. API Contract

/search

- Following enhancements are included:
 - **Order preparation time**
 - Logistics buyer NP can send order preparation time in /search, which can be used by LSP to optimize scheduling of their task allocation algorithm;
 - Order preparation time = max(time to ship as per catalog, time when RTS expected to be sent);

```
{
  "context":
  {
    "domain": "nic2004:60232",
    "country": "IND",
    "city": "std:08010",
    "action": "search",
    "core_version": "1.2.0",
    "bap_id": "logistics_buyer.com",
    "bap_uri": "https://logistics_buyer.com/ondc",
    "transaction_id": "T1",
    "message_id": "M1",
    "timestamp": "2023-06-06T21:00:00.000Z",
    "ttl": "PT30S"
  },
  "message":
  {
    "intent":
    {
      "category":
      {
        "id": "Immediate Delivery11"
      },
      "provider":
      {
        "time12":
        {
          "days": "1,2,3,4,5,6,713",
          "schedule":
          {
            "holidays14":
            [
```

¹⁰ [City code](#) should match the city for fulfillment start location;

¹¹ enum:

"Express Delivery" - ship by air;

"Standard Delivery" - ship by surface

- "Immediate Delivery" is for S2D <= 60 min;
- "Same Day Delivery"
- "Next Day Delivery"

If id is "Standard Delivery":

- all possible options should be returned in /on_search (including for "Immediate Delivery", "Same Day Delivery", "Next Day Delivery");

If id is "Express Delivery":

- all possible options should be returned in /on_search;

¹² store pickup timings;

¹³ days of week: 1 - Monday till 7 - Sunday;

¹⁴ list of future dated holidays;

```

    "2023-06-29",
    "2023-08-15"
  ],
  "duration15": "PT30M",
  "range16": {
    "start": "1100",
    "end": "2100"
  }
},
"fulfillment": {
  "type": "Delivery17",
  "start": {
    "location": {
      "gps": "12.453544,77.92837918",
      "address": {
        "area_code": "560041"
      }
    }
  },
  "authorization19": {
    "type": "OTP20"
  }
},
"end": {
  "location": {
    "gps": "12.453544,77.92837921",
    "address": {
      "area_code": "560001"
    }
  }
},
"authorization22": {
  "type": "OTP23"
}
},
"payment": {

```

¹⁵ order preparation time, indicates approx timing for logistics ready_to_ship;

¹⁶ **define store order pickup timing, for "days" above, here it means the store order pickup timings are from 1100 to 2100 from Monday to Sunday; end time must be greater than start time and (end time - start time) must be less than 24 hrs;**

¹⁷ **enum - "Delivery", "Return";**

mapping from old to new enum - ("CoD", "Prepaid" => "Delivery"), "Reverse QC" => "Return";

¹⁸ **minimum 6 digit decimal precision;**

¹⁹ **optional, only required if logistics buyer NP wants pickup authorization; if LSP doesn't support authorization, they should not respond to the search request;**

²⁰ **enum - "OTP";**

²¹ **minimum 6 digit decimal precision;**

²² **optional, only required if logistics buyer NP wants delivery authorization; if LSP doesn't support authorization, they should not respond to the search request;**

²³ **enum - "OTP";**


```

    "type": "POST-FULFILLMENT"24,
    "@ondc/org/collection_amount"25: "300.00"
  },
  "@ondc/org/payload_details": {
    "weight": {
      "unit": "kilogram"26,
      "value": 1
    },
    "dimensions"27: {
      "length": {
        "unit": "centimeter"28,
        "value": 1
      },
      "breadth": {
        "unit": "centimeter",
        "value": 1
      },
      "height": {
        "unit": "centimeter",
        "value": 1
      }
    },
    "category"29: "Grocery",
    "value": {
      "currency": "INR",
      "value": "300.00"
    },
    "dangerous_goods"30: false
  }
}
}
}

```

N.B.

1. LSP does a serviceability check based on the fulfillment start & end locations and provides a quote for all fulfillment options that meet the search intent;
2. LSP should support search for parent categories such as "Standard Delivery", "Express Delivery" or child categories such as "Immediate Delivery", "Same Day Delivery", "Next Day Delivery". Usage of these categories is defined above;
3. LSP should return all catalog options for "Standard Delivery" including for child categories such as "Immediate Delivery", "Same Day Delivery", "Next Day Delivery". LSP should also return all catalog options for "Express Delivery";

²⁴ enum - "ON-ORDER" (prepaid), "ON-FULFILLMENT" (CoD), "POST-FULFILLMENT" (post fulfillment);

²⁵ required for type = "ON-FULFILLMENT";

²⁶ enum - "kilogram", "gram" (aligns with this [list](#));

²⁷ package dimensions - mandatory for intercity shipments, optional for hyperlocal;

²⁸ enum [here](#);

²⁹ use category from [here](#) (e.g. Grocery, Fashion, etc);

³⁰ optional, set to true when payload includes hazardous goods;

/on_search

- Following enhancements are included:
 - **Motorable distance, average time to pickup (useful for Immediate Delivery)**
 - LSP can provide motorable distance (preferably **OSRM**), average time to pickup as part of the catalog response;
 - Motorable distance will enable logistics buyer NP to optimize their rate card for disconnected logistics;
 - Average time to pickup (1st mile) will enable logistics buyer NP to correctly estimate their O2D TAT for the buyer;
 - **Catalog response should be sent directly to logistics buyer NP;**

```
{
  "context":
  {
    "domain": "nic2004:60232",
    "country": "IND",
    "city": "std:080",
    "action": "on_search",
    "core_version": "1.2.0",
    "bap_id": "logistics_buyer.com",
    "bap_uri": "https://logistics_buyer.com/ondc",
    "bpp_id": "lsp.com",
    "bpp_uri": "https://lsp.com/ondc",
    "transaction_id": "T1",
    "message_id": "M1",
    "timestamp": "2023-06-06T21:00:30.000Z"
  },
  "message":
  {
    "catalog":
    {
      "bpp/descriptor":
      {
        "name31": "LSP Aggregator Inc",
        "tags":
        [
          {
            "code": "bpp_terms",
            "list":
            [
              {
                "code": "static_terms",
                "value": ""
              },
              {
                "code": "static_terms_new",
                "value": "https://github.com/ONDC-Official/NP-Static-Terms/lspNP_LSP/1.0/tc.pdf"
              },
              {
                "code": "effective_date",
                "value": "2023-10-01T00:00:00.000Z"
              }
            ]
          }
        ]
      }
    }
  },
}
```

³¹ name of logistics aggregator or logistics provider, as applicable;

```
"bpp/providers":
[
{
  "id": "P1",
  "descriptor":
  {
    "name": "LSP Courier Inc",
    "short_desc": "LSP Courier Inc",
    "long_desc": "LSP Courier Inc"
  },
  "categories":
  [
    {
      "id": "Immediate Delivery",
      "time32":
      {
        "label": "TAT",
        "duration": "PT60M",
        "timestamp33": "2023-06-06"
      }
    }
  ],
  "fulfillments":
  [
    {
      "id": "1",
      "type": "Delivery",
      "start":
      {
        "time":
        {
          "duration34": "PT15M"
        }
      },
      "tags":
      [
        {
          "code": "distance35",
          "list":
          [
            {
              "code": "motorable_distance_type",
              "value": "kilometer36"
            },
            {
              "code": "motorable_distance37",
              "value": "1.8"
            }
          ]
        }
      ]
    }
  ],
  {
    "id": "2",
    "type": "RTO"
  }
]
```

³² category level TAT for S2D (ship-to-delivery), can be overridden by item-level TAT whenever there are multiple options for the same category (e.g. 30 min, 45 min, 60 min, etc.);

³³ refers to date (vis-a-vis Context.timestamp) for which this TAT is provided (same day for Immediate Delivery / SDD, next day for NDD, appropriate date for other categories);

³⁴ average time to pickup (ISO8601 Duration);

³⁵ useful for P2P (hyperlocal);

³⁶ enum - "mile", "kilometer", "meter";

³⁷ shortest (preferably OSRM) distance between start & end locations;

```

    }
  ],
  "locations"38:
  [
    {
      "id": "L1",
      "gps": "12.967555,77.749666",
      "address":
      {
        "street": "Jayanagar 4th Block",
        "city": "Bengaluru",
        "area_code": "560076",
        "state": "KA"
      }
    }
  ],
  "items":
  [
    {
      "id": "I1",
      "parent_item_id": "",
      "category_id": "Immediate Delivery",
      "fulfillment_id": "1"39,
      "descriptor":
      {
        "code": "P2P"40,
        "name": "60 min delivery",
        "short_desc": "60 min delivery for F&B",
        "long_desc": "60 min delivery for F&B"
      },
      "price":
      {
        "currency": "INR",
        "value": "59.00"41
      },
      "time"42:
      {
        "label": "TAT",
        "duration": "PT45M",
        "timestamp"43: "2023-06-06"
      }
    },
    {
      "id": "I2",
      "parent_item_id"44: "I1",
      "category_id": "Immediate Delivery",
      "fulfillment_id": "2"45,
      "descriptor":
      {
        "code": "P2P",
        "name": "RTO quote",
        "short_desc": "RTO quote",

```

³⁸ mandatory only for cases where shipment has to be dropped off at LSP location; **not required for P2P**;

³⁹ indicates forward shipment item with quote;

⁴⁰ type of shipment; enum - "P2P" (point-to-point) and "P2H2P" (point-to-hub-to-point). P2H2P fulfillments require different packaging and AWB no. This is required for merchants to decide on the packaging required for shipment;

⁴¹ **price is tax inclusive here, itemized in /on_init**;

⁴² optional, item level TAT will override category-level TAT, if specified;

⁴³ **refers to date (vis-a-vis Context.timestamp) for which this TAT is provided (same day for Immediate Delivery / SDD, next day for NDD, appropriate date for other categories) - required for P2P**;

⁴⁴ RTO quote linked to forward shipment quote through parent_item_id;

⁴⁵ indicates RTO item with quote;

```

    "long_desc": "RTO quote"
  },
  "price": {
    "currency": "INR",
    "value": "23.60"46
  },
  "time"47: {
    "label": "TAT",
    "duration": "PT60M",
    "timestamp"48: "2023-06-06"
  }
}
]
}
}
}
}

```

Example

1. Logistics buyer NP can send /search with category "Standard Delivery" or "Express Delivery" and the LSP is required to respond with all appropriate options e.g. for "Standard Delivery", the LSP responds with options for "Immediate Delivery", "Same Day Delivery", "Next Day Delivery", as applicable;
2. Logistics buyer NP can also send /search for specific category such as "Immediate Delivery", "Same Day Delivery", "Next Day Delivery" and the LSP either responds with options available or doesn't respond;
3. Catalog options provided by LSP can be either "P2P" (point-to-point) or "P2H2P" (point-to-hub-to-point);
4. For "P2P", the rider delivers the package directly from the pickup location (merchant's) to the drop-off location (buyer's), for "P2H2P", the package is routed through a hub;
5. Logistics buyer NP can inform the merchants of the type of packaging required based on the catalog option i.e. "P2P" or "P2H2P". Since "P2H2P" fulfillments are routed through a hub, AWB no is required;
6. For different logistics categories, following validations should be in place:
 - a. "Immediate Delivery" - time.duration <= PT60M for the same day (time.timestamp is same day as Context.timestamp);
 - b. "Same Day Delivery" - for the same day (time.timestamp is same day as Context.timestamp);
 - c. "Next Day Delivery" - for the next day (time.timestamp is the day after the day for Context.timestamp)

⁴⁶ RTO quote (tax inclusive), will be added when RTO triggered through /on_cancel;

⁴⁷ optional, item level TAT will override category-level TAT, if specified;

⁴⁸ refers to date (vis-a-vis Context.timestamp) for which this TAT is provided (same day for Immediate Delivery / SDD, next day for NDD, appropriate date for other categories);

/on_search sync response (ACK)

In the sync response to /on_search, the logistics buyer NP can provide the following information:

- Acceptance or rejection of LSP static terms

/on_search (ACK response)

```
"response":
{
  "context":
  {
    "domain": "nic2004:60232",
    "country": "IND",
    "city": "std:080",
    "action": "on_search",
    "core_version": "1.2.0",
    "bap_id": "logistics_buyer.com",
    "bap_uri": "https://logistics_buyer.com/ondc",
    "bpp_id": "lsp.com",
    "bpp_uri": "https://lsp.com/ondc",
    "transaction_id": "T1",
    "message_id": "M1",
    "timestamp": "2023-06-06T21:00:30.000Z"
  },
  "message":
  {
    "ack":
    {
      "status": "ACK",
      "tags":
      [
        {
          "code": "bap_terms",
          "list":
          [
            {
              "code": "accept_bpp_terms",
              "value": "Y49"
            }
          ]
        }
      ]
    }
  }
}
```

⁴⁹ enum - "Y" (yes), "N" (no);

/init

```
{
  "context":
  {
    "domain": "nic2004:60232",
    "country": "IND",
    "city": "std:080",
    "action": "init",
    "core_version": "1.2.0",
    "bap_id": "logistics_buyer.com",
    "bap_uri": "https://logistics_buyer.com/ondc",
    "bpp_id": "lsp.com",
    "bpp_uri": "https://lsp.com/ondc",
    "transaction_id": "T1",
    "message_id": "M2",
    "timestamp": "2023-06-06T21:30:00.000Z",
    "ttl": "PT30S"
  },
  "message":
  {
    "order":
    {
      "provider":
      {
        "id": "P1",
        "locations50":
        [
          {
            "id": "L1"
          }
        ]
      },
      "items":
      [
        {
          "id": "I1",
          "fulfillment_id": "1",
          "category_id": "Immediate Delivery",
          "descriptor":
          {
            "code": "P2P"
          }
        }
      ],
      "fulfillments":
      [
        {
          "id": "1"51,
          "type": "Delivery",
          "start":
          {
            "location":
            {
              "gps": "12.453544,77.928379",
              "address52":
              {
                "name": "My store name",
                "building": "My building name",

```

⁵⁰ mandatory only if provider.locations was returned in /on_search;

⁵¹ fulfillment_id from /on_search corresponding to appropriate type;

⁵² **name + building + locality < 190 chars, name != locality;**

```

        "locality": "My street name",
        "city": "Bengaluru",
        "state": "Karnataka",
        "country": "India",
        "area_code": "560041"
    },
    "contact": {
        "phone": "9886098860",
        "email": "abcd.efgh@gmail.com"
    },
    "end": {
        "location": {
            "gps": "12.453544,77.928379",
            "address53: {
                "name": "My house #",
                "building": "My house or building name",
                "locality": "My street name",
                "city": "Bengaluru",
                "state": "Karnataka",
                "country": "India",
                "area_code": "560076"
            }
        },
        "contact": {
            "phone": "9886098860",
            "email54": "abcd.efgh@gmail.com"
        }
    },
    "billing55: {
        "name56": "ONDC Logistics Buyer NP",
        "address57: {
            "name": "My house or building no",
            "building": "My house or building name",
            "locality": "Jayanagar",
            "city": "Bengaluru",
            "state": "Karnataka",
            "country": "India",
            "area_code": "560076"
        },
        "tax_number58": "XXXXXXXXXXXXXXXXXX",
        "phone": "9886098860",
        "email59": "abcd.efgh@gmail.com",
        "created_at": "2023-02-06T21:30:00.000Z",
        "updated_at": "2023-02-06T21:30:00.000Z"
    },

```

⁵³ name + building + locality < 190 chars, name != locality;

⁵⁴ optional

⁵⁵ billing details of the marketplace (MSN or ISN retail seller app) for invoicing;

⁵⁶ name on the invoice;

⁵⁷ address on invoice;

⁵⁸ required - GST no for logistics buyer NP;

⁵⁹ required;


```
"payment"60:
{
  "@ondc/org/collection_amount"61:"300.00",
  "collected_by"62:"BPP",
  "type":"ON-FULFILLMENT",
  "@ondc/org/settlement_details"63:
  [
    {
      "settlement_counterparty":"buyer-app",
      "settlement_type":"upi"64,
      "beneficiary_name":"xxxxx",
      "upi_address":"gft@oksbi",
      "settlement_bank_account_no":"XXXXXXXXXX",
      "settlement_ifsc_code":"XXXXXXXXXX"
    }
  ]
}
}
```

N.B.

1. LSP is required to perform serviceability check while processing /init and return appropriate [error codes](#) if the pickup and / or dropoff locations aren't serviceable;

⁶⁰ assumes LSP is collecting COD payment for the order;

⁶¹ optional - only for ON-FULFILLMENT (CoD);

⁶² required for type ON-ORDER, ON-FULFILLMENT;

⁶³ optional - to be provided by LBNP if they're not collecting payment;

⁶⁴ enum - "upi", "neft", "rtgs"; for "upi", upi_address needs to be entered while for "neft", "rtgs", settlement_bank_account_no, settlement_ifsc_code needs to be entered;

/on_init

- Following enhancements are included:
 - **Cancellation terms**
 - Cancellation terms will be defined by the LSP for the order in /on_init;
 - Cancellation terms will be defined based on the fulfillment state & reason codes and include cancellation fees (if any);
 - If LSP cancellation terms are not acceptable to Logistics Buyer NP, they should NACK the /on_init with error code 62505;
 - **Itemizing tax in quote**
 - Currently, fulfillment charges provided by LSP are tax inclusive;
 - With this change, fulfillment charges will have separate quote line item for taxes;
 - Logistics buyer NP can forward the itemized fulfillment charges to retail buyer NP by consolidating the tax line items for other fulfillment charges;
 - **Inline check for rider availability (hyperlocal only)**
 - LSP can provide information on whether inline check for rider availability was done;
 - **Change in quote title type**
 - Change in quote title type to align with retail spec; updated enum & mapping is defined below:

<u>Current</u>	<u>New</u>	<u>Meaning</u>
Delivery Charge	delivery	Delivery charge
RTO Charge	rto	RTO charge
Reverse QC Charge	reverseqc	Reverse QC charge
Tax	tax	Tax on delivery / rto / reverseqc
	diff	Additional delivery charge due to weight differential
	tax_diff	Tax on additional delivery charge due to weight differential
	discount	Discount on delivery charges (if any)

- Configurable terms for [transaction level contract](#);

```
{
  "context":
  {
    "domain": "nic2004:60232",
    "country": "IND",
    "city": "std:080",
    "action": "on_init",
    "core_version": "1.2.0",
    "bap_id": "logistics_buyer.com",
    "bap_uri": "https://logistics_buyer.com/ondc",
  }
}
```

```

"bpp_id": "lsp.com",
"bpp_uri": "https://lsp.com/ondc",
"transaction_id": "T1",
"message_id": "M2",
"timestamp": "2023-02-06T21:30:30.000Z"
},
"message":
{
  "order":
  {
    "provider":
    {
      "id": "P1",
      "locations65":
      [
        {
          "id": "L1"
        }
      ]
    },
    "items":
    [
      {
        "id": "I1",
        "fulfillment_id": "1"
      }
    ]
  },
  "fulfillments":
  [
    {
      "id": "1",
      "type": "Delivery",
      "start":
      {
        "location":
        {
          "gps": "12.4535445,77.9283792",
          "address":
          {
            "name": "Store name",
            "building": "House or building name",
            "locality": "Locality",
            "city": "Bengaluru",
            "state": "Karnataka",
            "country": "India",
            "area_code": "560041"
          }
        }
      },
      "contact":
      {
        "phone": "9886098860",
        "email": "abcd.efgh@gmail.com"
      }
    },
    "end":
    {
      "location":
      {
        "gps": "12.453544,77.928379",
        "address":
        {
          "name": "My house #",

```

⁶⁵ mandatory only if provider.locations was returned in /on_search;

```

        "building": "My house or building name",
        "locality": "locality",
        "city": "Bengaluru",
        "state": "Karnataka",
        "country": "India",
        "area_code": "560076"
    },
    "contact": {
        "phone": "9886098860",
        "email": "abcd.efgh@gmail.com"
    },
    "tags": [
        {
            "code": "rider_check66",
            "list": [
                {
                    "code": "inline_check_for_rider",
                    "value": "yes"
                }
            ]
        }
    ]
},
"quote": {
    "price": {
        "currency": "INR",
        "value": "59.00"
    },
    "breakup": [
        {
            "@ondc/org/item_id": "I1",
            "@ondc/org/title_type": "delivery67",
            "price": {
                "currency": "INR",
                "value": "50.00"
            }
        },
        {
            "@ondc/org/item_id": "I1",
            "@ondc/org/title_type": "tax",
            "price": {
                "currency": "INR",
                "value": "9.00"
            }
        }
    ]
},
"ttl": "PT15M68"

```

⁶⁶ check for rider availability, will be useful for Immediate Delivery;

⁶⁷ enum - delivery, tax, rto, diff;
mapping from old to new enum - ("Delivery Charge", "Reverse QC Charge" => "delivery"; "RTO Charge"
-> "rto"; "Tax" => "tax";

⁶⁸ ttl for quote;

```

},
"payment":
{
  "@ondc/org/collection_amount69":"300.00",
  "type":"ON-FULFILLMENT70",
  "collected_by":"BPP",
  "@ondc/org/settlement_details71":
  [
    {
      "settlement_counterparty":"buyer-app",
      "settlement_type":"upi",
      "beneficiary_name":"xxxxx",
      "upi_address":"gft@oksbi",
      "settlement_bank_account_no":"XXXXXXXXXXXX",
      "settlement_ifsc_code":"XXXXXXXXXX"
    }
  ]
},
"cancellation_terms":
[
  {
    "fulfillment_state":
    {
      "descriptor":
      {
        "code":"Pending72",
        "short_desc73":"008"
      }
    },
    "cancellation_fee74":
    {
      "percentage":"0.00",
      "amount":
      {
        "currency":"INR",
        "value":"0.00"
      }
    }
  },
  {
    "fulfillment_state":
    {
      "descriptor":
      {
        "code":"Agent-assigned",
        "short_desc":"001,003"
      }
    },
    "cancellation_fee":
    {
      "percentage":"100.00",
      "amount":
      {
        "currency":"INR",

```

⁶⁹ optional - only for ON-FULFILLMENT (CoD);

⁷⁰ enum - "ON-FULFILLMENT" (CoD), "POST-FULFILLMENT" (post fulfillment billing), "ON-ORDER" (prepaid, typically thru wallet);

⁷¹ optional

⁷² fulfillment state (can be wildcard "*", for all states);

⁷³ reason code (can be wildcard "*", for all reason codes);

⁷⁴ cancellation fee, if any, for this state & reason code - either percentage (of order value minus taxes, in the above format between "0.00" and "100.00") or amount needs to be provided (if both provided, minimum should be considered by LBNP); LSP can calculate based on what needs to be deducted e.g. delivery charges, etc;

```

      "value": "50.00"
    }
  },
  {
    "fulfillment_state":
    {
      "descriptor":
      {
        "code": "Order-picked-up",
        "short_desc": "001,003"
      }
    },
    "cancellation_fee":
    {
      "percentage": "100.00",
      "amount":
      {
        "currency": "INR",
        "value": "50.00"
      }
    }
  },
  {
    "fulfillment_state":
    {
      "descriptor":
      {
        "code": "Out-for-delivery",
        "short_desc": "011,012,013,014,01575"
      }
    },
    "cancellation_fee":
    {
      "percentage": "100.00",
      "amount76:
      {
        "currency": "INR",
        "value": "50.00"
      }
    }
  }
],
"tags":
[
  {
    "code": "bpp_terms77",
    "list":
    [
      {
        "code": "max_liability",
        "value": "2"
      },
      {
        "code": "max_liability_cap",
        "value": "10000"
      }
    ]
  }
]

```

⁷⁵ list of reason codes delimited by "," (above shows RTO reason codes where buyer is not available);

⁷⁶ in case of RTO where buyer is responsible, the forward & reverse shipment costs may be included in the cancellation fees;

⁷⁷ logistics buyer NP has to accept LSP terms in /confirm. If logistics buyer doesn't accept these terms, they should NACK /on_init with error code 62501;

```

    "code": "mandatory_arbitration",
    "value": "false"
  },
  {
    "code": "court_jurisdiction",
    "value": "Bengaluru"
  },
  {
    "code": "delay_interest",
    "value": "1000"
  },
  {
    "code": "static_terms"78,
    "value": "https://github.com/ONDC-Official/protocol-network-extension/discussions/79"
  }
]
}
}
}

```

N.B.

1. LSP is required to perform check for serviceability, for the /init request, and return appropriate error codes along with the response;
2. Order quote should comply with the following:
 - a. Precision for all prices in quote can be maximum of 2 decimal digits;
 - b. Order value, i.e. `quote.price.value` = Σ quote line items, i.e. `quote.breakup[].price.value`;
 - c. Quote provided here should match what was provided in /on_search and is frozen until order /confirm;

⁷⁸ may be replaced with the URL of standard terms & conditions of network participant;

/confirm

- Following enhancements are included:
 - **Fulfillment state TAT breach**
 - LSP provides a ship-to-delivery (S2D) TAT to the logistics buyer NP, along with a quote;
 - Retail Seller NP uses this to compute an order-to-delivery (O2D) TAT as follows: ship-to-delivery (S2D) + order-to-ship (O2S) + average time to pickup (proposed above);
 - S2D TAT will also be a part of the order in /confirm and /on_confirm;
 - When there is an O2D breach, the retail buyer NP / seller NP may initiate cancellation of the retail order / fulfillment which will cascade into cancellation of the logistics order;
 - Liabilities in case of O2D breach will be defined in the terms & conditions;
 - **RTO flow**
 - Every fulfillment will have an identifier for whether RTO results in actual return to origin;
 - **Codifying reverse QC SOP**
 - Reverse QC SOP is in the form of an online checklist, with or without instructions, and provided by Seller NP;
 - Logistics agent updates the online checklist with status of inspection of return items;
 - Sample checklist is included [here](#) and this should preferably include instructions on whether a item (for return) can be picked up or not;
 - **Changing tag structure for RTS**
 - tag structure for ready_to_ship will be changed to ensure consistency with the new tag structure;

```
{
  "context":
  {
    "domain":"nic2004:60232",
    "country":"IND",
    "city":"std:080",
    "action":"confirm",
    "core_version":"1.2.0",
    "bap_id":"logistics_buyer.com",
    "bap_uri":"https://logistics_buyer.com/ondc",
    "bpp_id":"lsp.com",
    "bpp_uri":"https://lsp.com/ondc",
    "transaction_id":"T1",
    "message_id":"M3",
    "timestamp":"2023-06-06T22:00:00.000Z",
    "ttl":"PT30S"
  },
  "message":
  {
    "order":
    {
      "id":"O279",
      "state":"Created",

```

⁷⁹ alphanumeric, up to 32 characters;


```

"provider":
{
  "id": "P1",
  "locations80":
  [
    {
      "id": "L1"
    }
  ]
},
"items":
[
  {
    "id": "I1",
    "fulfillment_id": "1",
    "category_id": "Immediate Delivery",
    "descriptor":
    {
      "code": "P2P"
    },
    "time81":
    {
      "label": "TAT",
      "duration": "PT45M",
      "timestamp": "2023-06-06"
    }
  }
],
"quote":
{
  "price":
  {
    "currency": "INR",
    "value": "59.00"
  },
  "breakup":
  [
    {
      "@ondc/org/item_id": "I1",
      "@ondc/org/title_type": "delivery",
      "price":
      {
        "currency": "INR",
        "value": "50.00"
      }
    },
    {
      "@ondc/org/item_id": "I1",
      "@ondc/org/title_type": "tax",
      "price":
      {
        "currency": "INR",
        "value": "9.00"
      }
    }
  ]
},
"fulfillments":
[

```

⁸⁰ mandatory only if provider.locations was returned in /on_search;

⁸¹ S2D TAT; if S2D TAT and / or average pickup time is different from what was quoted earlier in /on_search, LSP can NACK /confirm with error code 60008;

```
{
  "id": "1",
  "type": "Delivery",
  "@ondc/org/awb_no82": "1227262193237777",
  "start": {
    "time83": {
      "duration": "PT15M"
    },
    "person": {
      "name": "person_name"
    },
    "location": {
      "gps": "12.4535445,77.9283792",
      "address": {
        "name": "Store name",
        "building": "House or building name",
        "locality": "Locality",
        "city": "Bengaluru",
        "state": "Karnataka",
        "country": "India",
        "area_code": "560041"
      }
    },
    "contact": {
      "phone": "9886098860",
      "email": "abcd.efgh@gmail.com"
    },
    "instructions84": {
      "code": "285",
      "short_desc": "value of PCC86",
      "long_desc": "additional instructions for pickup87",
      "additional_desc88": {
        "content_type": "text/html",
        "url": "https://reverse_qc_sop_form.htm"
      }
    }
  },
  "end": {
    "person": {
      "name": "person_name"
    },
    "location": {
```

⁸² AWB # can be provided by logistics buyer NP (in /confirm or /update) or LSP (in /on_confirm or /on_update) and is mandatory only if item.descriptor.code = "P2H2P". Can be 11-16 digits;

⁸³ average pickup time;

⁸⁴ "code", "short_desc" required if "ready_to_ship" = yes in /confirm;

⁸⁵ type of PCC: enum - "2" - merchant order no, "3" - other pickup confirmation code, "4" - OTP;

⁸⁶ value of PCC:

merchant order no - max 6 alphanumeric;

other pickup confirmation code - max 6 alphanumeric;

OTP - max 6 alphanumeric;

⁸⁷ optional

⁸⁸ (optional) reverse QC online checklist, may be provided for fulfillment type = "Return";

```
{
  "gps": "12.453544,77.928379",
  "address": {
    "name": "My house #",
    "building": "My house or building name",
    "locality": "locality",
    "city": "Bengaluru",
    "state": "Karnataka",
    "country": "India",
    "area_code": "560076"
  },
  "contact": {
    "phone": "9886098860",
    "email": "abcd.efgh@gmail.com"
  },
  "instructions89": {
    "code": "390",
    "short_desc": "value of DCC91",
    "long_desc": "additional instructions for delivery92"
  },
  "tags": [
    {
      "code": "state",
      "list": [
        {
          "code": "ready_to_ship93",
          "value": "no94"
        }
      ]
    },
    {
      "code": "rto_action",
      "list": [
        {
          "code": "return_to_origin95",
          "value": "no"
        }
      ]
    },
    {
      "code": "reverseqc_input96",
      "list": [

```

⁸⁹ optional - may be provided after order is picked up;

⁹⁰ type of DCC: enum - "1" - OTP, "2" - other DCC, "3" - no delivery code;

⁹¹ value of DCC (required for code 1 & 2):

other DCC - max 6 alphanumeric;

OTP - max 6 alphanumeric;

⁹² optional

⁹³ replaces "@ondc/org/order_ready_to_ship";

⁹⁴ enum - "yes", "no";

⁹⁵ whether RTO results in actual return to origin;

⁹⁶ may be provided if fulfillment.type="Return";

codified reverse qc checklist (includes product attributes & questionnaire);

universal checklist is [here](#);

```

    {
      "code": "P001",
      "value": "Atta"
    },
    {
      "code": "P003",
      "value": "1"
    },
    {
      "code": "Q001",
      "value": ""
    }
  ]
}
],
"billing":
{
  "name": "ONDC Logistics Buyer NP",
  "address":
  {
    "name": "name",
    "building": "My house or building name",
    "locality": "My street name",
    "city": "Bengaluru",
    "state": "Karnataka",
    "country": "India",
    "area_code": "560076"
  },
  "tax_number": "29AAACU1901H1ZK",
  "phone": "9886098860",
  "email": "abcd.efgh@gmail.com",
  "created_at": "2023-06-06T21:30:00.000Z",
  "updated_at": "2023-06-06T21:30:00.000Z"
},
"payment":
{
  "@ondc/org/collection_amount": "300.00",
  "collected_by": "BPP",
  "type": "ON-FULFILLMENT",
  "@ondc/org/settlement_details97":
  [
    {
      "settlement_counterparty": "buyer-app",
      "settlement_type": "upi",
      "upi_address": "gft@oksbi",
      "settlement_bank_account_no": "XXXXXXXXXX",
      "settlement_ifsc_code": "XXXXXXXXXX"
    }
  ]
},
"@ondc/org/linked_order":
{
  "items":
  [
    {
      "category_id": "Grocery98",
      "descriptor":
      {
        "name": "Atta"
      }
    }
  ]
}

```

⁹⁷ optional

⁹⁸ use category from [here](#);

```

    },
    "quantity":
    {
      "count":2,
      "measure":
      {
        "unit":"kilogram99",
        "value":0.5
      }
    },
    "price":
    {
      "currency":"INR",
      "value":"150.00"
    }
  }
],
"provider":
{
  "descriptor":
  {
    "name":"Aadishwar Store"
  },
  "address":
  {
    "name":"KHB Towers",
    "building":"Building or House No",
    "locality":"Koramangala",
    "city":"Bengaluru",
    "state":"Karnataka",
    "area_code":"560070"
  }
},
"order":
{
  "id":"O1",
  "weight":
  {
    "unit":"kilogram100",
    "value":1
  },
  "dimensions101":
  {
    "length":
    {
      "unit":"centimeter102",
      "value":1
    },
    "breadth":
    {
      "unit":"centimeter",
      "value":1
    },
    "height":
    {
      "unit":"centimeter",
      "value":1
    }
  }
}

```

⁹⁹ unit weight, enum [here](#);

¹⁰⁰ total weight, enum [here](#);

¹⁰¹ (mandatory for intercity, optional for hyperlocal) if weight and / or dimensions provided are different from what was initially provided (in /search), LSP can NACK /confirm with error code 60011;

¹⁰² enum [here](#);

```

    }
  },
  "tags":
  [
    {
      "code": "bpp_terms",
      "list":
      [
        {
          "code": "max_liability",
          "value": "2"
        },
        {
          "code": "max_liability_cap",
          "value": "10000"
        },
        {
          "code": "mandatory_arbitration",
          "value": "false"
        },
        {
          "code": "court_jurisdiction",
          "value": "Bengaluru"
        },
        {
          "code": "delay_interest",
          "value": "1000"
        },
        {
          "code": "static_terms",
          "value": "https://github.com/ONDC-Official/protocol-network-extension/discussions/79"
        }
      ]
    },
    {
      "code": "bap_terms",
      "list":
      [
        {
          "code": "accept_bpp_terms103",
          "value": "Y"
        }
      ]
    }
  ],
  "created_at": "2023-06-06T22:00:00.000Z",
  "updated_at": "2023-06-06T22:00:00.000Z"
}

```

N.B.

1. LSP is required to perform check for serviceability, for the /confirm request, and return appropriate error codes along with the response;
2. LSP creates the shipment manifest on successful processing of /confirm;

¹⁰³ logistics buyer NP must accept LSP terms. If not accepted, LSP can NACK /confirm with error code 65002;

/on_confirm

```
{
  "context":
  {
    "domain": "nic2004:60232",
    "country": "IND",
    "city": "std:080",
    "action": "on_confirm",
    "core_version": "1.2.0",
    "bap_id": "logistics_buyer.com",
    "bap_uri": "https://logistics_buyer.com/ondc",
    "bpp_id": "lsp.com",
    "bpp_uri": "https://lsp.com/ondc",
    "transaction_id": "T1",
    "message_id": "M3",
    "timestamp": "2023-06-06T22:00:30.000Z"
  },
  "message":
  {
    "order":
    {
      "id": "O2",
      "state": "Accepted104",
      "provider":
      {
        "id": "P1",
        "locations105":
        [
          {
            "id": "L1"
          }
        ]
      },
    },
    "items":
    [
      {
        "id": "I1",
        "fulfillment_id": "1",
        "category_id": "Same Day Delivery",
        "descriptor":
        {
          "code": "P2P"
        },
        "time106":
        {
          "label": "TAT",
          "duration": "PT45M",
          "timestamp": "2023-06-06"
        }
      }
    ],
    "quote":
    {
      "price":
      {
        "currency": "INR",
        "value": "59.00"
      }
    }
  }
}
```

¹⁰⁴ enum - "Created", "Accepted", "Cancelled", LSP may accept order after "ready_to_ship" from LBNP;

¹⁰⁵ required only if in /confirm;

¹⁰⁶ S2D TAT; If S2D TAT and / or average pickup time is different from what was quoted earlier in /on_search, Logistics Buyer NP can NACK /on_confirm with error code 62506;

```

},
"breakup":
[
  {
    "@ondc/org/item_id": "I1",
    "@ondc/org/title_type": "delivery",
    "price":
    {
      "currency": "INR",
      "value": "50.00"
    }
  },
  {
    "@ondc/org/item_id": "I1",
    "@ondc/org/title_type": "tax",
    "price":
    {
      "currency": "INR",
      "value": "9.00"
    }
  }
]
},
"fulfillments":
[
  {
    "id": "1",
    "type": "Delivery",
    "state":
    {
      "descriptor":
      {
        "code": "Pending"
      }
    }
  },
  "@ondc/org/awb_no107": "122726219323777",
  "tracking": false108,
  "start109":
  {
    "person":
    {
      "name": "person_name"
    },
    "location":
    {
      "gps": "12.453544,77.928379",
      "address":
      {
        "name": "Store name",
        "building": "My house or building name",
        "locality": "My street name",
        "city": "Bengaluru",
        "state": "Karnataka",
        "country": "India",
        "area_code": "560041"
      }
    }
  },
  "contact":
  {

```

¹⁰⁷ required only if item.descriptor.code = "P2H2P"; can be provided by logistics buyer (/confirm or /update) or LSP (/on_confirm or /on_update);

¹⁰⁸ should be true if live tracking is enabled;

¹⁰⁹ required if "ready_to_ship"="yes" in /confirm;


```

"phone": "9886098860",
"email": "abcd.efgh@gmail.com"
},
"instructions":
{
  "code": "2",
  "short_desc": "value of PCC",
  "long_desc": "QR code will be attached to package",
  "images":
  [
    "link to downloadable shipping label (required for P2H2P)110"
  ],
  "additional_desc111":
  {
    "content_type": "text/html",
    "url": "https://reverse_qc_sop_form.htm"
  }
},
"time":
{
  "duration": "PT15M",
  "range112":
  {
    "start": "2023-06-06T22:30:00.000Z",
    "end": "2023-06-06T22:45:00.000Z"
  }
},
"end":
{
  "person":
  {
    "name": "person_name"
  },
  "location":
  {
    "gps": "12.4535445,77.9283792",
    "address":
    {
      "name": "My house or building #",
      "building": "My house or building name",
      "locality": "My street name",
      "city": "Bengaluru",
      "state": "Karnataka",
      "country": "India",
      "area_code": "560076"
    }
  },
  "contact":
  {
    "phone": "9886098860",
    "email": "abcd.efgh@gmail.com"
  },
  "instructions":
  {
    "code": "3",
    "short_desc": "value of DCC"
  },
  "time":

```

¹¹⁰ only if "ready_to_ship" = "yes" in /confirm;

¹¹¹ optional

¹¹² in response to RTS, LSP provides a pickup time slot. This time slot could be specific in case of "Immediate Delivery" or could be the entire day for SDD / NDD etc;

```
{
  "range"113:
    {
      "start": "2023-06-06T23:00:00.000Z",
      "end": "2023-06-06T23:15:00.000Z"
    }
  },
  "agent"114:
    {
      "name": "agent_name",
      "phone": "9886098860"
    },
  "vehicle"115:
    {
      "registration": "3LVJ945"
    },
  "tags":
    [
      {
        "code": "weather_check"116,
        "list":
          [
            {
              "code": "raining",
              "value": "yes"
            }
          ]
      },
      {
        "code": "state",
        "list":
          [
            {
              "code": "ready_to_ship",
              "value": "yes"
            }
          ]
      },
      {
        "code": "rto_action",
        "list":
          [
            {
              "code": "return_to_origin",
              "value": "no"
            }
          ]
      },
      {
        "code": "reverseqc_input"117,
        "list":
          [
            {
              "code": "P001",
              "value": "Atta"
            }
          ]
      }
    ]
}
```

¹¹³ may be provided if "ready_to_ship"="yes" in /confirm;

¹¹⁴ optional, may be provided for hyperlocal, inter-city delivery for last mile shipment;

¹¹⁵ optional

¹¹⁶ **weather conditions that buyer can be made aware of, may be provided for P2P (hyperlocal);**

¹¹⁷ may be provided if fulfillment.type="Return";

codified reverse qc checklist (includes product attributes & questionnaire);

universal checklist is [here](#);

```

    },
    {
      "code": "P003",
      "value": "1"
    },
    {
      "code": "Q001",
      "value": ""
    }
  ]
}
],
"billing"118:
{
  "name": "ONDC Logistics Buyer NP",
  "address":
  {
    "name": "My building no",
    "building": "My building name",
    "locality": "My street name",
    "city": "Bengaluru",
    "state": "Karnataka",
    "country": "India",
    "area_code": "560076"
  },
  "tax_number": "XXXXXXXXXXXXXXXX",
  "phone": "9886098860",
  "email": "abcd.efgh@gmail.com",
  "created_at": "2023-02-06T21:30:00.000Z",
  "updated_at": "2023-02-06T21:30:00.000Z"
},
"payment"119:
{
  "@ondc/org/collection_amount": "300.00",
  "collected_by": "BPP",
  "type": "ON-FULFILLMENT",
  "@ondc/org/settlement_details"120:
  [
    {
      "settlement_counterparty": "buyer-app",
      "settlement_type": "upi",
      "upi_address": "gft@oksbi",
      "settlement_bank_account_no": "XXXXXXXXXX",
      "settlement_ifsc_code": "XXXXXXXXXX"
    }
  ]
},
"@ondc/org/linked_order":
{
  "items":
  [
    {
      "category_id": "Grocery"121,
      "descriptor":
      {
        "name": "Atta"
      }
    }
  ],

```

¹¹⁸ billing should be same as in /init

¹¹⁹ as in /confirm;

¹²⁰ optional

¹²¹ use category from [here](#);

```

    "quantity":
    {
      "count":2,
      "measure":
      {
        "unit":"kilogram122",
        "value":0.5
      }
    },
    "price":
    {
      "currency":"INR",
      "value":"150.00"
    }
  },
  "provider":
  {
    "descriptor":
    {
      "name":"Aadishwar Store"
    },
    "address":
    {
      "name":"KHB Towers",
      "building":"Building or House No",
      "locality":"Koramangala",
      "city":"Bengaluru",
      "state":"Karnataka",
      "area_code":"560070"
    }
  },
  "order":
  {
    "id":"O1",
    "weight":
    {
      "unit":"kilogram123",
      "value":1
    },
    "dimensions124":
    {
      "length":
      {
        "unit":"centimeter",
        "value":1
      },
      "breadth":
      {
        "unit":"centimeter",
        "value":1
      },
      "height":
      {
        "unit":"centimeter",
        "value":1
      }
    }
  }
}

```

¹²² unit weight, enum - "kilogram", "gram" (aligns with this [list](#));

¹²³ total weight, enum - "kilogram", "gram" (aligns with this [list](#));

¹²⁴ if weight and / or dimensions provided are different from what was initially provided (in /search), LSP can NACK /confirm with error code 60011;

```

    },
    "cancellation_terms125":
    [
      {
        "fulfillment_state":
        {
          "descriptor":
          {
            "code": "Pending",
            "short_desc": "008"
          }
        },
        "cancellation_fee":
        {
          "percentage": "0.00",
          "amount":
          {
            "currency": "INR",
            "value": "0.00"
          }
        }
      },
      {
        "fulfillment_state":
        {
          "descriptor":
          {
            "code": "Agent-assigned",
            "short_desc": "001,003"
          }
        },
        "cancellation_fee":
        {
          "percentage": "100.00",
          "amount":
          {
            "currency": "INR",
            "value": "50.00"
          }
        }
      },
      {
        "fulfillment_state":
        {
          "descriptor":
          {
            "code": "Order-picked-up",
            "short_desc": "001,003"
          }
        },
        "cancellation_fee":
        {
          "percentage": "100.00",
          "amount":
          {
            "currency": "INR",
            "value": "50.00"
          }
        }
      }
    ],
    {

```

¹²⁵ as proposed in /on_init; if different from what was proposed in /on_init, buyer NP can NACK with error code 62509;

```

"fulfillment_state":
{
  "descriptor":
  {
    "code": "Out-for-delivery",
    "short_desc": "011,012,013,014,015"
  }
},
"cancellation_fee":
{
  "percentage": "100.00",
  "amount":
  {
    "currency": "INR",
    "value": "50.00"
  }
}
},
"tags":
[
  {
    "code": "bpp_terms",
    "list":
    [
      {
        "code": "max_liability",
        "value": "2"
      },
      {
        "code": "max_liability_cap",
        "value": "10000"
      },
      {
        "code": "mandatory_arbitration",
        "value": "false"
      },
      {
        "code": "court_jurisdiction",
        "value": "Bengaluru"
      },
      {
        "code": "delay_interest",
        "value": "1000"
      },
      {
        "code": "static_terms",
        "value": "https://github.com/ONDC-Official/protocol-network-extension/discussions/79"
      }
    ]
  },
  {
    "code": "bap_terms",
    "list":
    [
      {
        "code": "accept_bpp_terms",
        "value": "Y"
      }
    ]
  }
]
],

```

```
"created_at"126:"2023-02-06T22:00:00.000Z",  
"updated_at":"2023-02-06T22:00:30.000Z"  
}  
}  
}
```

N.B.

1. /on_confirm returns the fulfillment slot, agent details, vehicle details, E-way bill no, as applicable, for inter-state shipments;

¹²⁶ should match the corresponding timestamp in /confirm;

Rules for order confirmation

1. /confirm and /on_confirm calls should be idempotent;
2. Logistics Buyer NP (LBNP) will create the order id which will be unique across the network. In response to /confirm from LBNP, LSP sends /on_confirm with the same order_id;
3. Logistics Buyer NP (LBNP) sends /confirm, LSP validates¹²⁷ order:
 - a. if validation successful, LSP responds with ACK and creates order;
 - b. if validation not successful, LSP responds with NACK with error code **66002 (order validation failure)**:
 - i. after LBNP receives NACK, they should cancel the order with [reason code 996 \(order cancelled because of order confirmation failure\)](#);
 - ii. if this order is passed in subsequent API calls (/confirm, /status, /cancel, /update, etc.), LSP returns NACK with error code 66004 (order not found);
 - c. LSP may also respond with NACK due to other internal errors (http 504, retrievable error code **66001**):
 - i. LBNP should retry¹²⁸ /confirm. If after the retry interval, LBNP doesn't receive ACK/NACK, they should cancel the order with reason code **996**;
 - d. if LBNP doesn't receive ACK/NACK, they should retry /confirm (same as above). If within the retry interval, LBNP doesn't receive ACK/NACK, they should cancel the order with reason code **996**;
4. LSP sends /on_confirm, LBNP validates¹²⁹ order:
 - a. if validation successful, LBNP responds with ACK;
 - b. if validation not successful, LBNP responds with NACK with error code **63002 (order validation failure)**:
 - i. after LSP receives NACK, they should cancel the order with reason code **997 (order cancelled because of order confirmation failure)** & push the status change to LBNP;
 - ii. if this order (with cancellation state & reason code) is passed in subsequent API calls (/on_confirm, /on_status, /on_cancel, /on_update), LBNP responds with ACK;
 - c. LBNP may also NACK due to internal errors (http 504/503, retrievable error code **63001**):
 - i. LSP should retry¹³⁰ /on_confirm. If after the retry interval, LSP doesn't receive ACK/NACK, they should cancel the order with reason code **997**;
 - ii. If order is passed in subsequent API calls within the retry window (/status, /track, /cancel, /update), LSP responds with current order state along with error code 66003 (order processing in progress);
 - iii. If order is passed in subsequent API calls after cancellation (/status, /cancel, /update), LSP sends response with order state as cancelled;

¹²⁷ validation includes - order object (items / quantity / quote / fulfillment) same as in /on_init;

¹²⁸ retry will have a cap for no of attempts over specific duration to be decided by BNP;

¹²⁹ validation includes - order object (id / items / quantity / quote) same as in /confirm;

¹³⁰ retry will have a cap for no of attempts over specific duration to be decided by SNP;

/update

```
{
  "context":
  {
    "domain": "nic2004:60232",
    "country": "IND",
    "city": "std:080",
    "action": "update",
    "core_version": "1.2.0",
    "bap_id": "logistics_buyer.com",
    "bap_uri": "https://logistics_buyer.com/ondc",
    "bpp_id": "lsp.com",
    "bpp_uri": "https://lsp.com/ondc",
    "transaction_id": "T1",
    "message_id": "M4",
    "timestamp": "2023-06-06T22:30:00.000Z",
    "ttl": "PT30S"
  },
  "message":
  {
    "update_target": "fulfillment",
    "order":
    {
      "id": "O2",
      "items":
      [
        {
          "id": "I1",
          "category_id": "Same Day Delivery",
          "descriptor":
          {
            "code": "P2P"
          }
        }
      ]
    },
    "fulfillments":
    [
      {
        "id": "1",
        "type": "Delivery",
        "@ondc/org/awb_no131": "1227262193237777",
        "start":
        {
          "instructions132":
          {
            "code": "2133",
            "short_desc": "value of PCC134",
            "long_desc": "additional instructions for pickup",
            "additional_desc135":
            {
              "content_type": "text/html",
              "url": "https://reverse_qc_sop_form.htm"
            }
          }
        }
      }
    ]
  }
}
```

¹³¹ required if item.descriptor.code = "P2H2P" (may be provided in /confirm or /update by logistics buyer NP or /on_confirm or /on_update by LSP);

¹³² "code", "short_desc" required if "ready_to_ship" = yes;

¹³³ type of PCC: enum - "2" - merchant order no, "3" - other pickup confirmation code, "4" - OTP;

¹³⁴ value of PCC:

merchant order no - max 6 alphanumeric;

other pickup confirmation code - max 6 alphanumeric;

OTP - max 6 alphanumeric;

¹³⁵ reverse QC online checklist required only for fulfillment type = "Return" (if not in /confirm);

```

    }
  },
  "authorization"136:
  {
    "type": "OTP"137,
    "token": "OTP code",
    "valid_from": "2023-06-06T12:00:00.000Z",
    "valid_to": "2023-06-06T14:00:00.000Z"
  }
},
"end":
{
  "instructions"138:
  {
    "code": "2"139,
    "short_desc": "value of DCC"140,
    "long_desc": "additional instructions for delivery"
  }
},
"tags":
[
  {
    "code": "state",
    "list":
    [
      {
        "code": "ready_to_ship",
        "value": "yes"
      }
    ]
  }
]
},
"@ondc/org/linked_order"141:
{
  "items":
  [
    {
      "category_id": "Grocery",
      "descriptor":
      {
        "name": "Atta"
      },
      "quantity":
      {
        "count": 2,
        "measure":
        {
          "unit": "kilogram",
          "value": 0.5
        }
      }
    },
  ],
  "price":

```

¹³⁶ optional

¹³⁷ enum - "OTP", others TBD;

¹³⁸ optional

¹³⁹ type of DCC: enum - "1" - OTP, "2" - other DCC, "3" - no delivery code;

¹⁴⁰ value of DCC:

OTP - max 6 alphanumeric;

other DCC - max 6 alphanumeric;

¹⁴¹ optional, required for updating linked order details e.g. in case of part return or cancel. If not provided here, the linked order as provided in /confirm will hold;

```
{
  "currency": "INR",
  "value": "150.00"
}
],
"order": {
  "id": "O1",
  "weight142143

```

N.B.

1. /update can be used for any of the following:
 - a. Notify LSP that retail order is ready to ship;
 - b. Provide updated delivery instructions to LSP;
 - c. Update linked order details in case of part return / cancel for retail order;
 - d. **Update authorization details for pickup / delivery;**

¹⁴² only if weight changed from /confirm;

¹⁴³ only if dimensions changed from /confirm;

/on_update

- Following enhancements are included:
 - **Handling weight differential**
 - If LSP determines that there is a weight difference, they can update the differential cost and the weight used in calculation of this cost in /on_update;
 - In case of RTO, the differential cost for RTO can be updated in /on_cancel when the RTO request is initiated;
 - If logistics buyer NP ACKs /on_update, it means they accept the differential cost proposed by the LSP. **In this case, the differential cost will be added to forward shipment cost & reflected in cancellation fees, defined in the order cancellation terms, for cases where the order is subsequently cancelled;**
 - If the logistics buyer NP doesn't accept the differential cost (in /on_update or /on_cancel APIs), they can NACK /on_update with error code 62504. In this case, logistics buyer NP will still be responsible for the forward costs as included in /confirm & /on_confirm and will need to abide by the terms & conditions, for weight differential, as agreed to with the LSP;

```
{
  "context":
  {
    "domain":"nic2004:60232",
    "country":"IND",
    "city":"std:080",
    "action":"on_update",
    "core_version":"1.2.0",
    "bap_id":"logistics_buyer.com",
    "bap_uri":"https://logistics_buyer.com/ondc",
    "bpp_id":"lsp.com",
    "bpp_uri":"https://lsp.com/ondc",
    "transaction_id":"T1",
    "message_id":"M4",
    "timestamp":"2023-06-06T22:30:30.000Z"
  },
  "message":
  {
    "order":
    {
      "id":"O2",
      "state":"In-progress",
      "provider":
      {
        "id":"P1",
        "locations144":
        [
          {
            "id":"L1"
          }
        ]
      },
    },
    "items":
    [
      {
        "id":"I1",
        "fulfillment_id":"1",

```

¹⁴⁴ required only if in /confirm;

```

"category_id": "Same Day Delivery",
"descriptor":
{
  "code": "P2P"
},
"time":
{
  "label": "TAT",
  "duration": "PT45M",
  "timestamp": "2023-06-06"
}
},
],
"quote":
{
  "price":
  {
    "currency": "INR",
    "value": "88.50"
  },
  "breakup":
  [
    {
      "@ondc/org/item_id": "I1",
      "@ondc/org/title_type": "delivery",
      "price":
      {
        "currency": "INR",
        "value": "50.0"
      }
    },
    {
      "@ondc/org/item_id": "I1",
      "@ondc/org/title_type": "tax",
      "price":
      {
        "currency": "INR",
        "value": "9.0"
      }
    },
    {
      "@ondc/org/item_id": "I1145",
      "@ondc/org/title_type": "diff",
      "price":
      {
        "currency": "INR",
        "value": "25.0"
      }
    },
    {
      "@ondc/org/item_id": "I1146",
      "@ondc/org/title_type": "tax_diff",
      "price":
      {
        "currency": "INR",
        "value": "4.5"
      }
    }
  ]
}
]

```

¹⁴⁵ same item id as original item id since this is an additional weight-differential cost on top of the quoted price for the logistics service;

¹⁴⁶ same item id as original item id since this is an tax on additional weight-differential cost for the logistics service;

```

},
"fulfillments":
[
{
  "id": "1",
  "type": "Delivery",
  "state":
  {
    "descriptor":
    {
      "code": "Order-picked-up"
    }
  },
  "@ondc/org/awb_no147": "1227262193237777",
  "tracking": false,
  "start":
  {
    "person":
    {
      "name": "Ramu"
    },
    "location":
    {
      "gps": "12.453544,77.928379",
      "address":
      {
        "name": "name",
        "building": "My house or building name",
        "locality": "My street name",
        "city": "Bengaluru",
        "state": "Karnataka",
        "country": "India",
        "area_code": "560041"
      }
    },
    "contact":
    {
      "phone": "9886098860",
      "email": "abcd.efgh@gmail.com"
    },
    "time":
    {
      "duration": "PT15M",
      "range":
      {
        "start": "2023-06-06T23:45:00.000Z",
        "end": "2023-06-07T00:00:00.000Z"
      },
      "timestamp": "2023-06-07T00:00:00.000Z"
    },
    "instructions":
    {
      "code": "2",
      "short_desc": "value of PCC",
      "long_desc": "additional instructions for pickup",
      "images":
      [
        "link to downloadable shipping label (required for P2H2P)",
        "https://lsp.com/pickup_image.png148",
        "https://lsp.com/rider_location.png149"
      ]
    }
  }
}
]

```

¹⁴⁷ required for item.descriptor.code = "P2H2P";

¹⁴⁸ proof of order picked up - optional;

¹⁴⁹ proof of order picked up - optional;

```

    ],
    "additional_desc150":
    {
        "content_type": "text/html",
        "url": "https://reverse_qc_sop_form.htm"
    }
},
"end":
{
    "person":
    {
        "name": "person_name"
    },
    "location":
    {
        "gps": "12.453544,77.928379",
        "address":
        {
            "name": "My house or building #",
            "building": "My house or building name",
            "locality": "My street name",
            "city": "Bengaluru",
            "state": "Karnataka",
            "country": "India",
            "area_code": "560076"
        }
    },
    "contact151":
    {
        "phone": "9886098860",
        "email": "abcd.efgh@gmail.com"
    },
    "instructions152":
    {
        "code": "3",
        "short_desc": "value of DCC",
        "long_desc": "additional instructions for delivery",
        "images":
        [
            "https://lsp.com/delivery_image.png",
            "https://lsp.com/rider_location.png"
        ]
    },
    "time":
    {
        "range":
        {
            "start": "2023-06-07T02:00:00.000Z",
            "end": "2023-06-07T02:15:00.000Z"
        }
    }
},
"agent153":
{
    "name": "person_name",
    "phone": "9886098860"
},
"vehicle":

```

¹⁵⁰ optional (may be provided for reverse QC)

¹⁵¹ optional, if available;

¹⁵² optional, if available;

¹⁵³ optional

```
{
  "registration": "3LVJ945"
},
"@ondc/org/ewaybillno154": "EBN1",
"@ondc/org/ebnexpirydate": "2023-06-30T12:00:00.000Z"
},
"billing": {
  "name": "ONDC sellerNP",
  "address": {
    "name": "My building no",
    "building": "My building name",
    "locality": "My street name",
    "city": "Bengaluru",
    "state": "Karnataka",
    "country": "India",
    "area_code": "560076"
  },
  "tax_number": "XXXXXXXXXXXXXXXXXX",
  "phone": "9886098860",
  "email": "abcd.efgh@gmail.com",
  "created_at": "2023-06-06T21:30:00.000Z",
  "updated_at": "2023-06-06T21:30:00.000Z"
},
"payment": {
  "@ondc/org/collection_amount": "300.00",
  "collected_by": "BPP",
  "type": "ON-FULFILLMENT",
  "@ondc/org/settlement_details": [
    {
      "settlement_counterparty": "buyer-app",
      "settlement_type": "upi",
      "upi_address": "gft@oksbi",
      "settlement_bank_account_no": "XXXXXXXXXX",
      "settlement_ifsc_code": "XXXXXXXXXX"
    }
  ]
},
"@ondc/org/linked_order": {
  "items": [
    {
      "category_id": "Grocery",
      "descriptor": {
        "name": "Atta"
      },
      "quantity": {
        "count": 2,
        "measure": {
          "unit": "kilogram",
          "value": 0.5
        }
      }
    }
  ],
  "price":
```

¹⁵⁴ EBN # & expiry date will be provided by LSP for inter-state shipments


```
{
  "currency": "INR",
  "value": "150.00"
}
],
"provider":
{
  "descriptor":
  {
    "name": "Aadishwar Store"
  },
  "address":
  {
    "name": "KHB Towers",
    "building": "Building or House No",
    "street": "6th Block",
    "locality": "Koramangala",
    "city": "Bengaluru",
    "state": "Karnataka",
    "area_code": "560070"
  }
},
"order":
{
  "id": "O1",
  "weight":
  {
    "unit": "kilogram",
    "value": 1
  },
  "dimensions":
  {
    "length":
    {
      "unit": "centimeter",
      "value": 1
    },
    "breadth":
    {
      "unit": "centimeter",
      "value": 1
    },
    "height":
    {
      "unit": "centimeter",
      "value": 1
    }
  }
},
"tags":
[
  {
    "code": "diff_dim155",
    "list":
    [
      {
        "code": "unit",
        "value": "centimeter"
      }
    ]
  }
]
```

¹⁵⁵ updated dimensions, only if there is difference in dimensions;

```

    "code": "length",
    "value": "1.5"
  },
  {
    "code": "breadth",
    "value": "1.5"
  },
  {
    "code": "height",
    "value": "1.5"
  }
]
},
{
  "code": "diff_weight156",
  "list":
  [
    {
      "code": "unit",
      "value": "kilogram"
    },
    {
      "code": "weight",
      "value": "1.5"
    }
  ]
},
{
  "code": "diff_proof157",
  "list":
  [
    {
      "code": "type",
      "value": "image"
    },
    {
      "code": "url",
      "value": "https://lsp.com/sorter/images1.png"
    }
  ]
},
],
"updated_at": "2023-06-07T23:00:30.000Z"
}
}
}

```

N.B.

1. Differential weight update:
 - a. If Logistics buyer NP ACKs /on_update, it means they accept the updated quote, including the weight differential cost;
 - b. If Logistics buyer NP doesn't accept the updated quote, including the weight differential cost, they can NACK /on_update (with error code 62504). In this case, the LSP should revert the proposed changes in quote, order weight & dimensions and subsequent action will be as per the terms & conditions agreed upon between the logistics buyer NP & LSP;

¹⁵⁶ updated weight, only if there is difference in weight;

¹⁵⁷ in case of updated dimensions and / or weight, the proof (images from sorter) can be added here;

/cancel

```
{
  "context":
  {
    "domain": "nic2004:60232",
    "country": "IND",
    "city": "std:080",
    "action": "cancel",
    "core_version": "1.2.0",
    "bap_id": "logistics_buyer.com",
    "bap_uri": "https://logistics_buyer.com/ondc",
    "bpp_id": "lsp.com",
    "bpp_uri": "https://lsp.com/ondc",
    "transaction_id": "T1",
    "message_id": "M5",
    "timestamp": "2023-06-06T23:00:00.000Z",
    "ttl": "PT30S"
  },
  "message":
  {
    "order_id": "O2",
    "cancellation_reason_id": "011"158
  }
}
```

N.B.

1. /cancel will be used by the logistics buyer to cancel the order. LSP can cancel the order through unsolicited /on_cancel;
2. If cancellation reason is invalid, LSP can NACK /cancel with error code 60009;
3. For cancellation due to TAT breach (cancellation_reason_id = "007"), if LSP determines that fulfillment TAT is not breached, they can NACK /cancel with error code 60010;

¹⁵⁸ cancellation reason codes for [hyperlocal](#) & [inter-city](#);

/on_cancel

```
{
  "context":
  {
    "domain": "nic2004:60232",
    "country": "IND",
    "city": "std:080",
    "action": "on_cancel",
    "core_version": "1.2.0",
    "bap_id": "logistics_buyer.com",
    "bap_uri": "https://logistics_buyer.com/ondc",
    "bpp_id": "lsp.com",
    "bpp_uri": "https://lsp.com/ondc",
    "transaction_id": "T1",
    "message_id": "M5",
    "timestamp": "2023-06-06T23:00:30.000Z"
  },
  "message":
  {
    "order":
    {
      "id": "O2",
      "state": "Cancelled",
      "cancellation":
      {
        "cancelled_by": "logistics_buyer.com",
        "reason":
        {
          "id": "011"
        }
      },
      "provider":
      {
        "id": "P1",
        "locations":
        [
          {
            "id": "L1"
          }
        ]
      },
      "items":
      [
        {
          "id": "I1",
          "fulfillment_id": "1",
          "category_id": "Same Day Delivery",
          "descriptor":
          {
            "code": "P2P"
          },
          "time":
          {
            "label": "TAT",
            "duration": "PT45M",
            "timestamp": "2023-06-06"
          }
        }
      ],
      "quote159":

```

¹⁵⁹ quote should reflect the updated value of logistics order (including weight differential cost & tax if applicable) which has to be paid by logistics buyer NP;

```
{
  "price":
  {
    "currency": "INR",
    "value": "1"
  },
  "breakup":
  [
    {
      "@ondc/org/item_id": "I1",
      "@ondc/org/title_type": "delivery",
      "price":
      {
        "currency": "INR",
        "value": "1"
      }
    },
    {
      "@ondc/org/item_id": "I1",
      "@ondc/org/title_type": "tax",
      "price":
      {
        "currency": "INR",
        "value": "1"
      }
    }
  ]
},
"fulfillments":
[
  {
    "id": "1",
    "type": "Delivery",
    "state":
    {
      "descriptor":
      {
        "code": "Cancelled"
      }
    },
    "@ondc/org/awb_no": "122726219323777160",
    "tracking": false,
    "start":
    {
      "person":
      {
        "name": "person_name"
      },
      "location":
      {
        "gps": "12.453544,77.928379",
        "address":
        {
          "name": "name",
          "building": "My house or building name",
          "locality": "My street name",
          "city": "Bengaluru",
          "state": "Karnataka",
          "country": "India",
          "area_code": "560041"
        }
      }
    }
  },
]
```

¹⁶⁰ required for P2H2P fulfillments;

```

"contact":
{
  "phone":"9886098860",
  "email":"abcd.efgh@gmail.com"
},
"instructions":
{
  "code":"2",
  "short_desc":"value of PCC",
  "long_desc":"QR code will be attached to package",
  "additional_desc161:
  {
    "content_type":"text/html",
    "url":"https://reverse_qc_sop_form.htm"
  }
},
"time":
{
  "duration":"PT15M",
  "range":
  {
    "start":"2023-06-06T22:30:00.000Z",
    "end":"2023-06-06T22:45:00.000Z"
  }
},
"end":
{
  "person":
  {
    "name":"person_name"
  },
  "location":
  {
    "gps":"12.453544,77.928379",
    "address":
    {
      "name":"My house or building #",
      "building":"My house or building name",
      "locality":"My street name",
      "city":"Bengaluru",
      "state":"Karnataka",
      "country":"India",
      "area_code":"560076"
    }
  },
  "contact":
  {
    "phone":"9886098860",
    "email":"abcd.efgh@gmail.com"
  },
  "instructions":
  {
    "code":"3",
    "short_desc":"value of DCC"
  },
  "time":
  {
    "range":
    {
      "start":"2023-06-06T23:00:00.000Z",
      "end":"2023-06-06T23:15:00.000Z"
    }
  }
}

```

¹⁶¹ optional

```

    }
  },
  "agent":
  {
    "name": "agent_name",
    "phone": "9886098860"
  },
  "vehicle":
  {
    "registration": "3LVJ945"
  },
  "tags":
  [
    {
      "code": "igm_request162",
      "list":
      [
        {
          "code": "id",
          "value": "Issue1"
        }
      ]
    },
    {
      "code": "precancel_state163",
      "list":
      [
        {
          "code": "fulfillment_state",
          "value": "Order-picked-up"
        },
        {
          "code": "updated_at",
          "value": "2023-06-06T23:15:00.000Z"
        }
      ]
    }
  ]
},
  "billing":
  {
    "name": "ONDC Logistics Buyer NP",
    "address":
    {
      "name": "My building no",
      "building": "My building name",
      "locality": "My street name",
      "city": "Bengaluru",
      "state": "Karnataka",
      "country": "India",
      "area_code": "560076"
    },
    "tax_number": "XXXXXXXXXXXXXXXX",
    "phone": "9886098860",
    "email": "abcd.efgh@gmail.com",
    "created_at": "2023-06-06T21:30:00.000Z",
    "updated_at": "2023-06-06T21:30:00.000Z"
  },

```

¹⁶² (optional) reference to issue related to cancel request raised in IGM, this will be added here only after issue resolution and returned in all post-order APIs like /on_status, /on_update, /on_cancel;

¹⁶³ pre-cancel state of fulfillment, i.e. state of fulfillment prior to cancellation;

```
"payment":
{
"@ondc/org/collection_amount":"300.00",
"collected_by":"BPP",
"type":"ON-FULFILLMENT",
"@ondc/org/settlement_details":
[
{
"settlement_counterparty":"buyer-app",
"settlement_type":"upi",
"upi_address":"gft@oksbi",
"settlement_bank_account_no":"XXXXXXXXXX",
"settlement_ifsc_code":"XXXXXXXXXX"
}
]
},
"@ondc/org/linked_order":
{
"items":
[
{
"category_id":"Grocery",
"descriptor":
{
"name":"Atta"
},
"quantity":
{
"count":2,
"measure":
{
"unit":"kilogram",
"value":0.5
}
},
"price":
{
"currency":"INR",
"value":"150.00"
}
}
],
"provider":
{
"descriptor":
{
"name":"Aadishwar Store"
},
"address":
{
"name":"KHB Towers",
"building":"Building or House No",
"locality":"Koramangala",
"city":"Bengaluru",
"state":"Karnataka",
"area_code":"560070"
}
},
"order":
{
"id":"O1",
"weight":
{
"unit":"kilogram",
```



```
    "value":1
  },
  "dimensions":
  {
    "length":
    {
      "unit":"centimeter",
      "value":1
    },
    "breadth":
    {
      "unit":"centimeter",
      "value":1
    },
    "height":
    {
      "unit":"centimeter",
      "value":1
    }
  }
},
"updated_at":"2023-06-06T22:00:30.000Z"
}
}
```

/on_cancel (RTO flow)

RTO will be initiated when the end buyer does not accept or cannot be located for the order / fulfillment to be delivered;

In case of RTO:

- Every fulfillment will have an identifier for whether RTO results in actual return to origin (in /confirm);
- RTO, with return, will result in package being returned to the point of pickup for now (will be amended to allow merchant to provide return location which may be different from point of pickup);
- Buyers are within their rights to RTO, if the fulfillment is delivered beyond the fulfillment TAT, thus constituting a TAT breach;
- In the event of RTO, without any breach of TAT, the Logistics Buyer NP may recover both the forward shipping cost as well as RTO costs from the retail buyer NP. In this case, the LSP may be expected to help establish that the retail buyer refused to accept or was not available / couldn't be contacted to accept delivery;

RTO process flow is as follows:

- LSP tries to deliver the order to the retail buyer (it is assumed that a certain number of retries are built in for all deliveries except immediate delivery). Typical number of retries is 3 as per industry standard;
- Retail buyer does not accept the order or is unreachable / unavailable at the mentioned shipping address on attempt to deliver the order (LSP may be expected to help substantiate that reasonable attempt was made to connect and deliver the order to the retail buyer at the specified address);
- LSP initiates cancellation of logistics order, through /on_cancel, and provides specific reason code as defined, for [hyperlocal](#) & [inter-city](#), and AWB no (for forward shipment, if applicable);
- /on_cancel with reason code (that triggers RTO as defined in the reason codes) will initiate RTO and result in the following:
 - addition of new fulfillment record for the logistics order, with fulfillment start location / time, fulfillment state set to "RTO-Initiated";
 - quote for RTO will be added to the original quote (for forward shipment), matching the quote provided in logistics /on_search;
 - order state will be set to "Cancelled";
- If the RTO request is valid (as determined by the reason code and above info provided), the logistics buyer NP has to accept the request. If LSP sends a different quote, from what was sent in /on_search, the logistics buyer NP may reject the RTO request (with NACK and error code 62503) and the RTO request has to be resent with the correct quote;

- /on_cancel will also result in cascaded retail /on_cancel request, for the fulfillment, and include the new fulfillment record for RTO, fulfillment start location / time, fulfillment state, order state and quote for RTO (if retail buyer has to pay for the RTO cost as per the cancellation reason code);
- After RTO delivery is completed, LSP updates fulfillment status for RTO fulfillment to "RTO-Delivered", fulfillment end time, etc. through /on_status and this is cascaded to the retail order;
- In case of RTO without return, fulfillment state will be updated to "RTO-Disposed";

```
{
  "context":
  {
    "domain": "nic2004:60232",
    "country": "IND",
    "city": "std:080",
    "action": "on_cancel",
    "core_version": "1.2.0",
    "bap_id": "logistics_buyer.com",
    "bap_uri": "https://logistics_buyer.com/ondc",
    "bpp_id": "lsp.com",
    "bpp_uri": "https://lsp.com/ondc",
    "transaction_id": "T1",
    "message_id": "M5",
    "timestamp": "2023-06-06T23:00:30.000Z"
  },
  "message":
  {
    "order":
    {
      "id": "O2",
      "state": "Cancelled",
      "cancellation":
      {
        "cancelled_by": "lsp.com",
        "reason":
        {
          "id": "013"
        }
      }
    },
    "provider":
    {
      "id": "P1",
      "locations":
      [
        {
          "id": "L1"
        }
      ]
    },
    "items":
    [
      {
        "id": "I1",
        "fulfillment_id": "1",
        "category_id": "Same Day Delivery",
        "descriptor":
        {
          "code": "P2P"
        }
      },
      "time":
    ]
  }
}
```

```
{
  "label": "TAT",
  "duration": "PT45M",
  "timestamp": "2023-06-06"
},
{
  "id": "I2",
  "fulfillment_id": "1-RTO",
  "category_id": "Same Day Delivery",
  "descriptor": {
    "code": "P2P"
  },
  "time": {
    "label": "TAT",
    "duration": "PT45M",
    "timestamp": "2023-06-06"
  }
},
],
"quote164": {
  "price": {
    "currency": "INR",
    "value": "82.60"
  },
  "breakup": [
    {
      "@ondc/org/item_id": "I1",
      "@ondc/org/title_type": "delivery",
      "price": {
        "currency": "INR",
        "value": "50.00"
      }
    },
    {
      "@ondc/org/item_id": "I1",
      "@ondc/org/title_type": "tax",
      "price": {
        "currency": "INR",
        "value": "9.00"
      }
    },
    {
      "@ondc/org/item_id": "I2",
      "@ondc/org/title_type": "rto",
      "price": {
        "currency": "INR",
        "value": "20.0"165
      }
    }
  ],
  {
    "@ondc/org/item_id": "I2",
```

¹⁶⁴ quote should reflect the updated value of logistics order (including weight differential cost & tax if applicable for forward shipment & RTO) which has to be paid by logistics buyer NP;

¹⁶⁵ additional RTO charges;

```

"@ondc/org/title_type":"tax",
"price":
{
  "currency":"INR",
  "value":"3.60166"
}
},
"fulfillments":
[
{
  "id":"1",
  "type":"Delivery",
  "state":
  {
    "descriptor":
    {
      "code":"Pending"
    }
  },
  "@ondc/org/awb_no":"1227262193237777",
  "tracking":false,
  "start":
  {
    "person":
    {
      "name":"person_name"
    },
    "location":
    {
      "gps":"12.453544,77.928379",
      "address":
      {
        "name":"name",
        "building":"My house or building name",
        "locality":"My street name",
        "city":"Bengaluru",
        "state":"Karnataka",
        "country":"India",
        "area_code":"560041"
      }
    },
    "contact":
    {
      "phone":"9886098860",
      "email":"abcd.efgh@gmail.com"
    },
    "instructions":
    {
      "code":"2",
      "short_desc":"value of PCC",
      "long_desc":"QR code will be attached to package",
      "additional_desc":
      {
        "content_type":"text/html",
        "url":"https://reverse_qc_sop_form.htm"
      }
    },
    "time":
    {
      "duration":"PT15M",

```

¹⁶⁶ additional tax for RTO charges;

```

"range":
{
  "start":"2023-06-06T22:30:00.000Z",
  "end":"2023-06-06T22:45:00.000Z"
}
},
"end":
{
  "person":
  {
    "name":"person_name"
  },
  "location":
  {
    "gps":"12.4535445,77.9283792",
    "address":
    {
      "name":"My house or building #",
      "building":"My house or building name",
      "locality":"My street name",
      "city":"Bengaluru",
      "state":"Karnataka",
      "country":"India",
      "area_code":"560076"
    }
  },
  "contact":
  {
    "phone":"9886098860",
    "email":"abcd.efgh@gmail.com"
  },
  "instructions":
  {
    "code":"3",
    "short_desc":"value of DCC"
  },
  "time":
  {
    "range":
    {
      "start":"2023-06-06T23:00:00.000Z",
      "end":"2023-06-06T23:15:00.000Z"
    }
  },
  "agent":
  {
    "name":"agent_name",
    "phone":"9886098860"
  },
  "vehicle":
  {
    "registration":"3LVJ945"
  },
  "@ondc/org/ewaybillno":"EBN1",
  "@ondc/org/ebnextpirydate":"2023-06-30T12:00:00.000Z",
  "tags":
  [
    {
      "code":"rto_event",
      "list":
      [
        {

```

```

    "code": "retry_count",
    "value": "3"167
  },
  {
    "code": "rto_id",
    "value": "F1-RTO"168
  },
  {
    "code": "cancellation_reason_id",
    "value": "013"169
  },
  {
    "code": "sub_reason_id",
    "value": "004"170
  },
  {
    "code": "cancelled_by",
    "value": "lsp.com"171
  }
],
},
{
  "code": "igm_request"172,
  "list":
  [
    {
      "code": "id",
      "value": "Issue1"
    }
  ]
},
{
  "code": "precancel_state"173,
  "list":
  [
    {
      "code": "fulfillment_state",
      "value": "Order-picked-up"
    },
    {
      "code": "updated_at",
      "value": "2023-06-06T23:15:00.000Z"
    }
  ]
}
],
},
{
  "id": "1-RTO",
  "type": "RTO",
  "state":
  {
    "descriptor":
    {

```

¹⁶⁷ no of retries attempted;

¹⁶⁸ fulfillment id of RTO fulfillment;

¹⁶⁹ cancellation reason code from 2b [here](#);

¹⁷⁰ optional, may be provided in some cases e.g. if cancellation_reason_id is "008", sub_reason_id can include the pickup failure codes from [here](#);

¹⁷¹ bpp id of LSP;

¹⁷² (optional) reference to issue related to cancel request raised in IGM, this will be added here only after issue resolution and returned in all post-order APIs like /on_status, /on_update, /on_cancel;

¹⁷³ pre-cancel state of fulfillment, i.e. state of fulfillment prior to cancellation;

```

    "code": "RTO-Initiated"174
  },
  "start":
  {
    "time":
    {
      "timestamp": "2023-06-07T14:00:00.000Z"
    }
  }
},
"billing":
{
  "name": "ONDC sellerNP",
  "address":
  {
    "name": "My building no",
    "building": "My building name",
    "locality": "My street name",
    "city": "Bengaluru",
    "state": "Karnataka",
    "country": "India",
    "area_code": "560076"
  },
  "tax_number": "XXXXXXXXXXXXXXXXXX",
  "phone": "9886098860",
  "email": "abcd.efgh@gmail.com",
  "created_at": "2023-06-06T21:30:00.000Z",
  "updated_at": "2023-06-06T21:30:00.000Z"
},
"payment":
{
  "@ondc/org/collection_amount": "300.00",
  "collected_by": "BPP",
  "type": "ON-FULFILLMENT",
  "@ondc/org/settlement_details":
  [
    {
      "settlement_counterparty": "buyer-app",
      "settlement_type": "upi",
      "upi_address": "gft@oksbi",
      "settlement_bank_account_no": "XXXXXXXXXXXX",
      "settlement_ifsc_code": "XXXXXXXXXX"
    }
  ]
},
"@ondc/org/linked_order":
{
  "items":
  [
    {
      "category_id": "Grocery",
      "descriptor":
      {
        "name": "Atta"
      },
      "quantity":
      {
        "count": 2,
        "measure":
        {

```

¹⁷⁴ enum - "RTO-Initiated", "RTO-Delivered", "RTO-Disposed";


```

        "unit": "kilogram",
        "value": 0.5
    },
    "price": {
        "currency": "INR",
        "value": "150.00"
    }
},
"provider": {
    "descriptor": {
        "name": "Aadishwar Store"
    },
    "address": {
        "name": "KHB Towers",
        "building": "Building or House No",
        "locality": "Koramangala",
        "city": "Bengaluru",
        "state": "Karnataka",
        "area_code": "560070"
    }
},
"order": {
    "id": "O1",
    "weight": {
        "unit": "kilogram",
        "value": 1
    },
    "dimensions": {
        "length": {
            "unit": "centimeter",
            "value": 1
        },
        "breadth": {
            "unit": "centimeter",
            "value": 1
        },
        "height": {
            "unit": "centimeter",
            "value": 1
        }
    }
},
"created_at": "2023-06-06T22:00:00.000Z",
"updated_at": "2023-06-06T22:00:30.000Z"
}
}
}

```

N.B.

1. In case of weight differential impacting RTO costs, LSP needs to provide the differential cost & updated weight / dimensions as in /on_update;
2. If Logistics buyer NP ACKs /on_cancel, it means they accept the updated quote, including the weight differential cost;
3. If Logistics buyer NP doesn't accept the updated quote, including the weight differential cost, they can NACK /on_cancel (with error code 62504). In this case, the LSP should revert the proposed changes in differential cost in the quote, updated order weight & dimensions and subsequent action will be as per the terms & conditions agreed upon between the logistics buyer NP & LSP;

/track

- Tracking may be enabled by retail buyer NPs for orders;
- To enable fulfillment tracking:
 - Tracking should be enabled for the fulfillment;
 - Fulfillment tracking becomes active after order is picked up for delivery and becomes inactive after the fulfillment is delivered;
 - /track request & /on_track response are cascaded from retail seller NP (logistics buyer NP) to LSP & back;

```
{
  "context":
  {
    "domain": "nic2004:60232",
    "action": "track",
    "country": "IND",
    "city": "std:080",
    "core_version": "1.2.0",
    "bap_id": "logistics_buyer.com",
    "bap_uri": "https://logistics_buyer.com/ondc",
    "bpp_id": "lsp.com",
    "bpp_uri": "https://lsp.com/ondc",
    "transaction_id": "T1",
    "message_id": "M6",
    "timestamp": "2023-06-06T23:30:00.000Z",
    "ttl": "PT30S"
  },
  "message":
  {
    "order_id": "O2"
  }
}
```

N.B.

1. Buyer NP will need to call /track to poll near real-time gps coordinates for the rider;
2. /track should be called only after the rider is assigned for the order. If /track is called for an order for which the order is not yet picked up or order has been picked up but fulfillment tracking is disabled, the LSP must return NACK with error code 60012;

/on_track

```
{
  "context":
  {
    "domain": "nic2004:60232",
    "action": "on_track",
    "country": "IND",
    "city": "std:080",
    "core_version": "1.2.0",
    "bap_id": "logistics_buyer.com",
    "bap_uri": "https://logistics_buyer.com/ondc",
    "bpp_id": "lsp.com",
    "bpp_uri": "https://lsp.com/ondc",
    "transaction_id": "T1",
    "message_id": "M6",
    "timestamp": "2023-06-06T23:30:30.000Z",
    "ttl": "PT30S"
  },
  "message":
  {
    "tracking":
    {
      "id": "F1175",
      "url": "https://lsp.com/ondc/track/F1176",
      "location":
      {
        "gps177": "12.974002,77.613458",
        "time":
        {
          "timestamp178": "2023-06-06T23:30:00.000Z"
        },
        "updated_at179": "2023-06-06T23:31:00.000Z"
      },
      "status180": "active",
      "tags181":
      [
        {
          "code": "order182",
          "list":
          [
            {
              "code": "id",
              "value": "O2"
            }
          ]
        },
        {
          "code": "config183",
```

¹⁷⁵ matches the fulfillment id for which tracking is enabled. In case multiple fulfillments have tracking enabled, the id should match the fulfillment which has been picked up;

¹⁷⁶ URL for non-hyperlocal tracking;

¹⁷⁷ gps coordinates of rider;

¹⁷⁸ time for which gps coordinates sent;

¹⁷⁹ time when gps coordinates updated;

¹⁸⁰ enum - "active", "inactive"; if inactive, "location" will be empty. If tracking is enabled, status should become "active" when order is picked up;

¹⁸¹ path as a sequence of gps lat/lng coordinates;

¹⁸² order id being tracked;

¹⁸³ whether tracking is by gps (point-to-point hyperlocal) or url (non-hyperlocal);

```

"list":
[
  {
    "code": "attr",
    "value": "tracking.location.gps"184
  },
  {
    "code": "type",
    "value": "live_poll"185
  }
]
},
{
  "code": "path",
  "list":
  [
    {
      "code": "lat_lng",
      "value": "12.974002,77.613458"
    },
    {
      "code": "sequence",
      "value": "1"
    }
  ]
},
{
  "code": "path",
  "list":
  [
    {
      "code": "lat_lng",
      "value": "12.974077,77.613600"
    },
    {
      "code": "sequence",
      "value": "2"
    }
  ]
},
{
  "code": "path",
  "list":
  [
    {
      "code": "lat_lng",
      "value": "12.974098,77.613699"
    },
    {
      "code": "sequence",
      "value": "3"
    }
  ]
}
]
}
}
}

```

¹⁸⁴ can be "tracking.location.gps", "tracking.url";

¹⁸⁵ enum - "live_poll" (for attr="tracking.location.gps"), "deferred" (for attr="tracking.url");

N.B.

1. To enable tracking:
 - Tracking should be enabled for the fulfillment;
 - Tracking becomes active after rider is assigned for the order and becomes inactive after the fulfillment is delivered;
 - /on_track will be solicited response as explained above;
 - /track request & /on_track response are cascaded from retail seller NP to LSP & back;

/status

```
{
  "context":
  {
    "domain": "nic2004:60232",
    "country": "IND",
    "city": "std:080",
    "action": "status",
    "core_version": "1.2.0",
    "bap_id": "logistics_buyer.com",
    "bap_uri": "https://logistics_buyer.com/ondc",
    "bpp_id": "lsp.com",
    "bpp_uri": "https://lsp.com/ondc",
    "transaction_id": "T1",
    "message_id": "M7",
    "timestamp": "2023-06-07T00:00:00.000Z",
    "ttl": "PT30S"
  },
  "message":
  {
    "order_id": "O2"
  }
}
```

/on_status

- Following enhancements are included:
 - **Proof of pickup / delivery**
 - Pickup & delivery are generally linked with pickup confirmation code (PCC) and delivery confirmation code (DCC);
 - In some cases, pickup & delivery can be authenticated using mechanisms such as OTP. In other cases, proof of pickup & delivery can optionally be communicated over the protocol;
 - Proof could include images (for pickup, delivery), last gps coordinates of rider, etc.;
 - Additional fulfillment states for hyperlocal, intercity;

```
{
  "context":
  {
    "domain": "nic2004:60232",
    "country": "IND",
    "city": "std:080",
    "action": "on_status",
    "core_version": "1.2.0",
    "bap_id": "logistics_buyer.com",
    "bap_uri": "https://logistics_buyer.com/ondc",
    "bpp_id": "lsp.com",
    "bpp_uri": "https://lsp.com/ondc",
    "transaction_id": "T1",
    "message_id": "M7",
    "timestamp": "2023-06-07T00:00:30.000Z"
  },
  "message":
  {
    "order":
    {
      "id": "O2",
      "state": "Completed"186,
      "cancellation"187:
      {
        "cancelled_by": "buyerNP.com",
        "reason":
        {
          "id": "011"
        }
      },
      "provider":
      {
        "id": "P1",
        "locations"188:
        [
          {
            "id": "L1"
          }
        ]
      },
      "items":
      [
        {
```

¹⁸⁶ enum - "Created", "Accepted", "In-progress", "Completed", "Cancelled";

¹⁸⁷ required if order state is "Cancelled";

¹⁸⁸ required only if in /confirm;


```

    "id": "I1",
    "fulfillment_id": "1",
    "category_id": "Same Day Delivery",
    "descriptor":
    {
      "code": "P2P"
    },
    "time":
    {
      "label": "TAT",
      "duration": "PT45M",
      "timestamp": "2023-06-06"
    }
  },
189
  {
    "id": "I2",
    "fulfillment_id": "1-RTO",
    "category_id": "Same Day Delivery",
    "descriptor":
    {
      "code": "P2P"
    },
    "time":
    {
      "label": "TAT",
      "duration": "PT45M",
      "timestamp": "2023-06-06"
    }
  }
],
"quote":
{
  "price":
  {
    "currency": "INR",
    "value": "108.50"
  },
  "breakup":
  [
    {
      "@ondc/org/item_id": "I1",
      "@ondc/org/title_type": "delivery",
      "price":
      {
        "currency": "INR",
        "value": "50.00"
      }
    },
    {
      "@ondc/org/item_id": "I1",
      "@ondc/org/title_type": "tax",
      "price":
      {
        "currency": "INR",
        "value": "9.00"
      }
    }
  ],
190
  {
    "@ondc/org/item_id": "I2",
    "@ondc/org/title_type": "rto",
    "price":

```

¹⁸⁹ if applicable;

¹⁹⁰ if applicable;

```
{
  "currency": "INR",
  "value": "20.00"
},
191{
  "@ondc/org/item_id": "I1",
  "@ondc/org/title_type": "diff",
  "price": {
    "currency": "INR",
    "value": "25.0"
  }
},
192{
  "@ondc/org/item_id": "I1",
  "@ondc/org/title_type": "tax_diff",
  "price": {
    "currency": "INR",
    "value": "4.5"
  }
}
],
"fulfillments": [
  {
    "id": "1",
    "type": "Delivery",
    "@ondc/org/awb_no": "1227262193237777",
    "state": {
      "descriptor": {
        "code": "Order-picked-up193",
        "short_desc": "pickup or delivery failed reason code194",
      }
    },
    "tracking": false,
    "start": {
      "person": {
        "name": "person_name"
      },
      "location": {
        "gps": "12.453544,77.928379",
        "address": {
          "name": "name",
          "building": "My house or building name",
          "locality": "My street name",
          "city": "Bengaluru",
          "state": "Karnataka",
          "country": "India",
          "area_code": "560041"
        }
      }
    }
  }
]
```

¹⁹¹ if applicable;

¹⁹² if applicable;

¹⁹³ enum for fulfillment state & mapping to order state for hyperlocal & inter-city is defined [here](#); enum for RTO - "RTO-Initiated", "RTO-Delivered", "RTO-Disposed";

¹⁹⁴ pickup failure reason code [here](#), delivery failure reason code [here](#);

```

    }
  },
  "contact":
  {
    "phone": "9886098860",
    "email": "abcd.efgh@gmail.com"
  },
  "time":
  {
    "duration": "PT15M",
    "range":
    {
      "start": "2023-06-06T22:30:00.000Z",
      "end": "2023-06-06T22:45:00.000Z"
    },
    "timestamp195": "2023-06-06T22:30:00.000Z"
  },
  "instructions196":
  {
    "code": "2",
    "name": "ONDC order",
    "short_desc": "value of PCC",
    "long_desc": "additional instructions for pickup197",
    "images198":
    [
      "link to downloadable shipping label",
      "https://lsp.com/pickup_image.png",
      "https://lsp.com/rider_location.png"
    ],
    "additional_desc199":
    {
      "content_type": "text/html",
      "url": "https://reverse_qc_sop_form.htm"
    }
  },
  "authorization200":
  {
    "type": "OTP201",
    "token": "OTP code",
    "valid_from": "2023-06-07T12:00:00.000Z",
    "valid_to": "2023-06-07T14:00:00.000Z"
  },
  "end":
  {
    "person":
    {
      "name": "person_name"
    },
    "location":

```

¹⁹⁵ required if fulfillment state code is "Order-picked-up", "Out-for-delivery", "Order-delivered"; after order is picked up, time range for delivery (end.time.range) may be updated, as applicable;

¹⁹⁶ optional

¹⁹⁷ optional

¹⁹⁸ images will have following entries in this order - shipping label (intercity) to be provided by LSP, proof of pickup (only after order picked up) - optional;

¹⁹⁹ (optional) if applicable, i.e. only in cases of reverse QC;

²⁰⁰ **optional, only for pickup authorization. If authorization is enabled by seller NP, LSP needs to provide authorization type & code, while updating fulfillment state to "Order-picked-up". Seller NP should verify this and for invalid / expired authorization, NACK the request with the following error codes:**

62510 - expired auth code (seller NP should regenerate the authorization code & send using /update);

62511 - invalid auth code;

²⁰¹ enum - "OTP", others TBD;

```
{
  "gps": "12.453544,77.928379",
  "address": {
    "name": "My house or building #",
    "building": "My house or building name",
    "locality": "My street name",
    "city": "Bengaluru",
    "state": "Karnataka",
    "country": "India",
    "area_code": "560076"
  },
  "contact": {
    "phone": "9886098860",
    "email": "abcd.efgh@gmail.com"
  },
  "time": {
    "range": {
      "start": "2023-06-06T23:00:00.000Z",
      "end": "2023-06-06T23:15:00.000Z"
    },
    "timestamp202": "2023-06-06T23:00:00.000Z"
  },
  "instructions203": {
    "code": "3",
    "name": "ONDC order",
    "short_desc": "value of DCC",
    "long_desc": "additional instructions for delivery204",
    "images205": [
      "https://lsp.com/delivery_image.png",
      "https://lsp.com/rider_location.png"
    ]
  },
  "authorization206": {
    "type": "OTP207",
    "token": "OTP code",
    "valid_from": "2023-06-07T12:00:00.000Z",
    "valid_to": "2023-06-07T14:00:00.000Z"
  },
  "agent208": {
    "name": "agent_name",
    "phone": "9886098860"
  },
}
```

²⁰² required if fulfillment state code is "Order-delivered";

²⁰³ optional

²⁰⁴ optional

²⁰⁵ proof of delivery - optional;

²⁰⁶ optional, only for delivery authorization. If authorization is enabled by seller NP, LSP needs to provide authorization type & code, while updating fulfillment state to "Order-delivered". Seller NP should verify this and for invalid / expired authorization, NACK the request with the following error codes:

62510 - expired auth;

62511 - invalid auth;

²⁰⁷ enum - "OTP", others TBD;

²⁰⁸ optional, may be provided for inter-city shipments for last mile delivery

```

"vehicle"209:
{
  "registration":"3LVJ945"
},
"@ondc/org/ewaybillno":"EBN1",
"@ondc/org/ebnexpirydate":"2023-06-30T12:00:00.000Z",
"tags":
[
  {
    "code":"reverseqc_output"210,
    "list":
    [
      {
        "code":"P001",
        "value":"Atta"
      },
      {
        "code":"P003",
        "value":"1"
      },
      {
        "code":"Q001",
        "value":"Y"211
      }
    ]
  },
  {
    "code":"fulfillment_delay"212,
    "list":
    [
      {
        "code":"state",
        "value":"Order-picked-up"213
      },
      {
        "code":"reason_id",
        "value":"002"214
      },
      {
        "code":"timestamp",
        "value":"2023-06-06T22:00:00.000Z"
      }
    ]
  },
  {
    "code":"tracking"215,
    "list":
    [
      {

```

²⁰⁹ optional

²¹⁰ may be provided if fulfillment.type="Return" for the following states - "Pickup-failed" or "Order-picked-up" or "Delivery-failed" or "Order-delivered";

codified reverse qc checklist (includes product attributes & questionnaire with answers);

universal checklist is [here](#);

²¹¹ enum - "Y" (yes) or "N" (no);

²¹² audit trail for pickup / delivery delay, including attempt made for pickup / delivery:

in case of pickup delay (with or without attempt), start time slot (start.time.range.start, start.time.range.end) needs to be updated and the pickup delay reason code and timestamp (when recorded) may be provided here;

in case of delivery delay (with or without attempt), end time slot (end.time.range.start, end.time.range.end) needs to be updated and the delivery delay reason code and timestamp (when recorded) may be provided here;

²¹³ enum - "Pickup-failed", "Order-picked-up", "Delivery-failed", "Order-delivered";

²¹⁴ reason codes for [pickup delay / failure](#), [delivery delay / failure](#);

²¹⁵ (optional) tracking attributes for fulfillment, may be provided, on or before fulfillment state of "Agent-assigned", if tracking = "true";

```

        "code": "gps_enabled"216,
        "value": "yes"
    },
    {
        "code": "url_enabled"217,
        "value": "no"
    },
    {
        "code": "url"218,
        "value": "https://sellerNP.com/ondc/tracking_url"
    }
]
},
{
    "id": "1-RTO",
    "type": "RTO",
    "state": {
        "descriptor": {
            "code": "RTO-Initiated"
        }
    },
    "start": {
        "time": {
            "range": {
                "start": "2023-06-06T23:00:00.000Z",
                "end": "2023-06-06T23:00:00.000Z"
            },
            "timestamp"219: "2023-06-06T23:00:00.000Z"
        }
    },
    "agent"220: {
        "name": "agent_name",
        "phone": "9886098860"
    }
},
"payment": {
    "@ondc/org/collection_amount": "300.00",
    "type": "POST-FULFILLMENT",
    "collected_by": "BPP",
    "status": "PAID",
    "time"221: {
        "timestamp": "2023-06-07T10:00:00.000Z"
    },
    "@ondc/org/settlement_details": [
        {

```

²¹⁶ whether tracking by GPS coordinates enabled;

²¹⁷ whether tracking by GPS coordinates enabled;

²¹⁸ if tracking by URL enabled, corresponding URL for tracking;

²¹⁹ required if fulfillment state code is "RTO-Initiated", "RTO-Delivered", "RTO-Disposed";

²²⁰ optional

²²¹ required for payment on delivery orders when order state is "Completed";

```

    "settlement_counterparty": "buyer-app",
    "settlement_type": "upi",
    "upi_address": "gft@oksbi",
    "settlement_bank_account_no": "XXXXXXXXXX",
    "settlement_ifsc_code": "XXXXXXXXXX",
    "settlement_status": "PAID",
    "settlement_reference": "XXXXXXXXXX",
    "settlement_timestamp": "2023-02-10T00:00:00.000Z"
  }
}
},
"billing":
{
  "name": "ONDC Seller NP",
  "address":
  {
    "name": "My building #",
    "building": "My building name",
    "locality": "My street name",
    "city": "Bengaluru",
    "state": "Karnataka",
    "country": "India",
    "area_code": "560076"
  },
  "tax_number": "XXXXXXXXXXXXXXXXXX",
  "phone": "9886098860",
  "email": "abcd.efgh@gmail.com"
},
"@ondc/org/linked_order":
{
  "items":
  [
    {
      "category_id": "Grocery",
      "descriptor":
      {
        "name": "Atta"
      },
      "quantity":
      {
        "count": 2,
        "measure":
        {
          "unit": "kilogram",
          "value": 0.5
        }
      },
      "price":
      {
        "currency": "INR",
        "value": "150.00"
      }
    }
  ],
  "provider":
  {
    "descriptor":
    {
      "name": "Aadishwar Store"
    },
    "address":
    {
      "name": "KHB Towers",
      "building": "Building or House No",

```

```

    "street": "6th Block",
    "locality": "Koramangala",
    "city": "Bengaluru",
    "state": "Karnataka",
    "area_code": "560070"
  },
  "order": {
    "id": "O1",
    "weight": {
      "unit": "kilogram",
      "value": 1
    },
    "dimensions": {
      "length": {
        "unit": "centimeter",
        "value": 1
      },
      "breadth": {
        "unit": "centimeter",
        "value": 1
      },
      "height": {
        "unit": "centimeter",
        "value": 1
      }
    }
  },
  "tags": [
    {
      "code": "diff_dim222",
      "list": [
        {
          "code": "unit",
          "value": "centimeter"
        },
        {
          "code": "length",
          "value": "1.5"
        },
        {
          "code": "breadth",
          "value": "1.5"
        },
        {
          "code": "height",
          "value": "1.5"
        }
      ]
    },
    {
      "code": "diff_weight223",

```

²²² updated dimensions, only if applicable;

²²³ updated weight, only if applicable;


```

"list":
[
  {
    "code": "unit",
    "value": "kilogram"
  },
  {
    "code": "weight",
    "value": "1.5"
  }
]
},
{
  "code": "diff_proof224",
  "list":
  [
    {
      "code": "type",
      "value": "image"
    },
    {
      "code": "url",
      "value": "https://lsp.com/sorter/images1.png"
    }
  ]
}
]
}
}
}

```

N.B.

1. /on_status response can be solicited or unsolicited;
2. /on_status returns the audit trail for order items (including status of item returns, cancels processed or in-progress), the current state of order items & fulfillments, order quote & breakup;
3. Buyer app can send NACK for /on_status with error code for cases such as:
 - a. NACK received by seller app for /on_confirm but seller app sends /on_status to update order state to "Accepted";
4. In case of NACK received for /on_status, seller app should rollback order to previous state of the order, i.e. the state prior to sending /on_status;

²²⁴ in case of updated dimensions and / or weight, the proof (images from sorter) can be added here, only if applicable;

Fulfillment states & mapping to order states

1. Hyperlocal (P2P)

Fulfillment State	When to assign state	Order state
"Pending"	default fulfillment state	"Created" or "Accepted"
"Searching-for-Agent"	after RTS	"In-progress"
"Agent-assigned"	rider assigned	"In-progress"
"At-pickup"²²⁵	rider reached pickup location	"In-progress"
"Order-picked-up"	order picked up by rider	"In-progress"
"Out-for-delivery"	Out for delivery	"In-progress"
"At-delivery"²²⁶	rider reached delivery location	"In-progress"
"Order-delivered"	delivered	"Completed"
"Cancelled"	cancelled	"Cancelled"

2. Intercity (P2H2P)

Fulfillment State	When to assign state	Order state
"Pending"	default fulfillment state	"Created" or "Accepted"
"Searching-for-Agent"	after RTS	"In-progress"
"Agent-assigned"	rider assigned	"In-progress"
"Out-for-pickup"	out for pickup	"In-progress"
"Pickup-failed"	pickup attempted but failed	"In-progress"
"Pickup-rescheduled"	pickup rescheduled	"In-progress"
"Order-picked-up"	picked up by rider	"In-progress"
"In-transit"	at source hub	"In-progress"
"At-destination-hub"	at destination hub	"In-progress"
"Out-for-delivery"	Out for delivery	"In-progress"
"Delivery-failed"	delivery attempted but failed	"In-progress"
"Delivery-rescheduled"	delivery rescheduled	"In-progress"
"Order-delivered"	delivered	"Completed"
"Cancelled"	cancelled	"Cancelled"

3. RTO

Fulfillment State	When to assign state	Order state
"RTO-Initiated"	When RTO has been initiated	Order state before RTO initiated

²²⁵ optional, to be set when rider reaches perimeter of pickup location, as specified by LSP;

²²⁶ optional, to be set when rider reaches perimeter of delivery location, as specified by LSP;

Fulfillment State	When to assign state	Order state
"RTO-Disposed"	RTO terminal state when "return to origin" not required	"Cancelled"
"RTO-Delivered"	RTO terminal state when "return to origin" required	"Cancelled"

Transaction level contract

Overview

1. Purpose of the transaction level contract between the participating NPs is explained [here](#) and includes the standard terms & conditions for the network participant(s);
2. The codified digital contract includes a set of configurable terms defined by the seller and a set of static terms which will be defined by the LSP;
3. A reference to these static terms will be a part of every online transaction between 2 participants;
4. Reference clauses to serve as a guide for participants to draft their static terms for the codified digital contract can be found [here](#);

Proposed flow

1. Static terms will be version-controlled in GitHub [repo](#):
 - a. All static terms will start with version 1.0.0 and subsequent changes will increment the version;
 - b. NP creates a PR for their static terms and ONDC will merge this PR and share the link with the NP:
 - i. NP forks the repository and creates a branch under the appropriate version, using the following naming convention:
 - <NP name>_role e.g. "Loadshare_LSP", "Shadowfax_LSP", etc.;
 - role can have following values - "LSP" (logistics service provider);
 - ii. NP merges their static terms into this branch;
 - iii. NP creates PR for these changes;
2. Static terms can be published by NP on multiple channels e.g. their website, ONDC portal, etc.
3. Communication & acceptance of static terms:
 - a. **Communication by LSP**
 - i. LSP sends the following additional info as part of /on_search:
 - **current** static terms link with version:
 - for the very 1st static terms, this will be a blank link;
 - **new** static terms link with version:
 - for the very 1st static terms, this will be the link for static terms in GitHub;
 - effective date for new static terms;
 - ii. Info in (i) above will be sent in /on_search for full catalog pull for 24 hrs, after this version of static terms is published;
 - iii. 24 hrs prior to expiry of effective date in (i) above, LSP resends additional info in (i) above;
 - iv. If LSP wants to change the effective date, they can update the effective date in (i) above and repeat (ii) & (iii);
 - v. If LSP wants to change their static terms, they need to resubmit their PR with new version no;
 - vi. It is recommended that the LSP should give at least 15 days of advance notice when they are changing their static terms;
 - b. **Acceptance by logistics buyer NP (BNP)**
 - i. If BNP accepts the static terms in /on_search, they need to send **accept="Y"** in ACK for /on_search;
 - ii. If BNP doesn't accept the static terms in /on_search, they need to send **accept="N"** in ACK for /on_search;
 - iii. If BNP doesn't accept the static terms for a LSP, they should stop sending order requests to the LSP;



- iv. Static terms agreed to between the BNP & LSP will be a part of /confirm and /on_confirm;